



Bachelor thesis

Department of Statistics

No 2026:x

Title in English

Title in Swedish

May 18, 2026

Charlotte Emily Iszatt & Milou Rossi

for the Alzheimer's Disease Neuroimaging Initiative*

*Data used in preparation of this article were obtained from the Alzheimer's Disease Neuroimaging Initiative (ADNI) database (adni.loni.usc.edu). As such, the investigators within the ADNI contributed to the design and implementation of ADNI and/or provided data but did not participate in the analysis or writing of this report. A complete listing of ADNI investigators can be found at:

http://adni.loni.usc.edu/wp-content/uploads/how_to_apply/ADNI_Acknowledgement_List.pdf

The abstract text can go here. The abstract text can go here. The
abstract text can go here. The abstract text can go here. The abstract
text can go here. The abstract text can go here. The abstract text can
go here. The abstract text can go here. The abstract text can go here.
The abstract text can go here. The abstract text can go here. The
abstract text can go here. The abstract text can go here. The abstract
text can go here. The abstract text can go here. The abstract text can
go here. Keywords: XXX, XXX, XXX

Data collection and sharing for the Alzheimer's Disease Neuroimaging Initiative (ADNI) is funded by the National Institute on Aging (National Institutes of Health Grant U19AG024904). The grantee organization is the Northern California Institute for Research and Education. In the past, ADNI has also received funding from the National Institute of Biomedical Imaging and Bioengineering, the Canadian Institutes of Health Research, and private sector contributions through the Foundation for the National Institutes of Health (FNIH) including generous contributions from the following: AbbVie, Alzheimer's Association; Alzheimer's Drug Discovery Foundation; Araclon Biotech; BioClinica, Inc.; Biogen; Bristol-Myers Squibb Company; CereSpir, Inc.; Cogstate; Eisai Inc.; Elan Pharmaceuticals, Inc.; Eli Lilly and Company; EuroImmun; F. Hoffmann-La Roche Ltd and its affiliated company Genentech, Inc.; Fujirebio; GE Healthcare; IXICO Ltd.; Janssen Alzheimer Immunotherapy Research & Development, LLC.; Johnson & Johnson Pharmaceutical Research & Development LLC.; Lumosity; Lundbeck; Merck & Co., Inc.; Meso Scale Diagnostics, LLC.; NeuroRx Research; Neurotrack Technologies; Novartis Pharmaceuticals Corporation; Pfizer Inc.; Piramal Imaging; Servier; Takeda Pharmaceutical Company; and Transition Therapeutics.

Table of contents

1	Introduction	5
1.1	Background	5
1.2	Research Context	6
1.3	Research Aims	7
2	Methodology	8
3	Statistical Theory	9
3.1	Survival Analysis	9
3.2	Cox Proportional Hazards Model	9
3.2.1	Model Assumptions	10
3.2.2	Concordance Index	11
3.3	Joint Modelling of Longitudinal and Survival Data	11
3.3.1	Linear Mixed-Effects Submodel	12
3.3.2	Survival Submodel	12
3.3.3	Model Assumptions	13
3.3.4	Estimation: Frequentist vs Bayesian Approach	13
3.3.5	The Gelman-Rubin Statistic	13
3.4	Joint Latent Class Mixed Model	15
3.4.1	Class Submodel	15
3.4.2	Model Assumptions	16
3.4.3	Estimation	16
3.4.4	Class Enumeration	17
3.5	Evaluation of Predictive Performance	18
3.5.1	Time-dependent AUC	18
3.5.2	Time-dependent Brier score	19
4	Data	20
4.1	Variables	21
4.1.1	Diagnosis status	21
4.1.2	p-tau181 and NfL	22
4.1.3	Age	23
4.1.4	Gender	23
4.1.5	APOE4-status	23
5	Results	23
5.1	Model Fit Overview	23
5.1.1	Cox PH Models	24
5.1.2	Joint Models	25
5.1.3	Joint Latent Class Mixed Models	28
5.2	Model Comparison	30

5.3	Biomarker Comparison	33
5.4	Sensitivity (or precision?) Analyses	36
6	Discussion	36
6.1	Main Findings and Conclusions	36
7	Bibliography	38
8	Appendix	42
8.1	Descriptive Statistics	42
8.2	Test Results	43
8.3	Residual Diagnostics	44
8.4	Convergence	45
8.5	Model Results	47
8.6	AUC and Brier Curves: Model Comparison	47
8.7	AUC and Brier Curves: Biomarker Comparison	48
8.8	Sensitivity analysis: 1+ Biomarker Measurements	48
8.9	Results of sensitivity analyses	48

Warning: package 'dplyr' was built under R version 4.4.3

Warning: package 'kableExtra' was built under R version 4.4.3

1 Introduction

1.1 Background

According to the World Health Organization (2025), Alzheimer's disease (AD) is the most common neurodegenerative disorder and a leading cause of dementia around the world with no definitive cure (WHO, 2025). The 2011 National Institute on Aging-Alzheimer's Association (NIA-AA) diagnostic guidelines describe the disease as a continuum with three stages: a symptom-free stage (pre-clinical AD), a pre-dementia stage (mild cognitive impairment (MCI) due to AD) and a dementia stage (dementia due to AD) (Carrillo et al. 2013, p. 596). Pathophysiological changes due to AD can begin up to 20 years before clinical symptoms appear, highlighting the crucial importance of early diagnosis and timely intervention to potentially slow the progression of the disease (Carrillo et al., p. 595).

In the context of AD, the term “biomarker” is commonly used and is defined as “a characteristic that is objectively measured and evaluated as an indicator of normal biological processes, pathogenic processes, or pharmacologic responses to a therapeutic intervention.” (Biomarkers Definitions Working Group, 2001, p. 91).

AD pathophysiology has typically been detected by positron emission topography (PET) to confirm the presence and degree of amyloid plaques in the brain or by quantification of biomarkers implicated in AD pathology in cerebrospinal fluid (CSF) (Mattsson et al., 2017, p. 557). The former method involves radiation and the latter requires a lumbar puncture in order to extract the CSF. WHO published a report in 2024 highlighting the promising progress made in recent years regarding the use of blood-based biomarkers to support AD diagnosis (WHO, p. 6). Specifically, recent breakthroughs in technology (Rissin et al., 2010) have allowed for quantification of biomarkers at much smaller concentrations than previous methods, typically being up to 1000 times more sensitive (Chang et al., 2012, p. 2). Using this method, biomarkers can be detected peripherally from their origin where concentrations are much smaller. These advances mean that more invasive procedures can be substituted with blood tests.

There are several core biomarkers used in the diagnosis and staging of AD (Jack et al., 2024), such as tau protein phosphorylated at the threonine-181 position (p-tau181), with concentrations expected to be higher in individuals with AD pathology (Wong et al., 2012, p. 815). Neurofilament light chain (NfL), a marker of neuronal injury not specific to AD (Jack et al., p. 5158), is also of importance. NfL is usually present inside neurons, and therefore concentrations are expected to be higher in individuals with neuronal injury symptomatic of AD (Clark

et al., 2021, p. 7). Research suggests that NfL may be used as a prognostic marker of cognitive decline which supports the diagnosis of a myriad of diseases, including frontotemporal dementia, whilst p-tau181 is more useful for differentiating AD from other neurodegenerative conditions (WHO, 2024, p. 7).

These two blood-based biomarkers, along with various other biomarkers and clinical measures related to AD, are the focus of the *Alzheimer’s Disease Neuroimaging Initiative* (ADNI), whose main goals are to provide open-access data to researchers worldwide and to improve the diagnosis of AD (ADNI, n.d.-a). Since 2004, researchers across more than 60 clinical sites in the USA and Canada have continuously collected and analyzed data throughout the different phases of the study. Blood samples from the ADNI cohort have been retrospectively analysed using the recently developed quantification method in order to measure p-tau181 and NfL concentrations (Zetterberg & Blennow, 2020; Blennow, 2018).

1.2 Research Context

Recent research has applied a range of statistical models to predict progression to dementia using biomarker data and evaluate their predictive accuracy. For instance, a 2021 study fitted a Cox proportional hazards (Cox PH) model incorporating baseline values of p-tau181, alongside demographic and genetic covariates such as age, sex and apolipoprotein- ϵ 4 (APOE- ϵ 4) status. The study included a total of 767 participants from the ADNI database. Predictive performance was evaluated using receiver operating characteristic (ROC) analysis at a fixed prediction horizon, yielding an area under the curve (AUC) of 0.78. This suggests that the model demonstrated good discriminative ability in identifying individuals who later developed dementia during follow-up (Wang et al., 2021, pp. 1-2).

Similarly, a large population-based study of 2148 older adults from Sweden applied a single multivariable Cox proportional hazards model including p-tau181, p-tau217, neurofilament light (NfL), and GFAP as simultaneous baseline predictors, to examine their predictive value for progression to all-cause and Alzheimer’s disease dementia. Predictive performance was assessed using a fixed 10-year prediction horizon, where the AUC was calculated based on dementia occurrence within 10 years. Within this framework, the presence of elevated values of one, two, or three biomarkers yielded AUC values of 74.3%, 77.0%, and 81.3% for AD-dementia, respectively, demonstrating progressively improved discriminative ability for identifying individuals who subsequently developed AD-dementia (Grande et al., 2025, pp. 2027-2031).

The joint modelling framework extends the Cox PH model by linking a longitudinal model which accounts for the effect of an endogenous time-dependent covariate measured with error, such as a biomarker (Rizopoulos, 2012, p. vii). This approach has been applied to model the relationship between repeated plasma p-tau181 and NfL measurements on the progression risk to AD (Yuan et al., 2024, p. 1). By evaluating AUC and Brier score at different time points and time spans, the researchers found that p-tau181 and NfL exhibited comparable performance in terms of discriminative capability and predictive accuracy for both 2-year

and 5-year prediction spans. In addition, it was concluded that combining the biomarker measurements in a multivariate joint model lead to improved performance (Yuan et al., 2024, p. 4).

The simultaneous modelling of multivariate longitudinal biomarkers and clinical endpoints when heterogeneity is highly suspected has been handled by joint latent class mixed models (JLCMMs) (Proust-Lima et al., 2023, p. 3996). JLCMMs model complex disease progression whilst offering a method for the exploration of new pathological hypotheses and identification of subphenotypes. One application of a multi-dimensional JLCMM captured heterogeneity in the trajectories of three cognitive domains (memory, language, and executive functioning) from the time of Alzheimer’s dementia diagnosis (Scollard et al., 2026, p. 383). They identified five latent classes differentiated by overall level of impairment (low-high) and rate of decline (slow-fast). Within each latent class, the pattern of decline was relatively consistent across cognitive domains, suggesting parallel deterioration across functions rather than domain-specific divergence. In addition, class membership was associated with several key variables, such as APOE- ϵ 4 status, sex, race, education and neuropathological burden (Scollard et al., 2026, p. 383). Despite these advances, such modeling approaches have rarely been applied to longitudinal plasma biomarkers in AD research.

Additionally, while the predictive performance of most of these modelling approaches has been examined separately, often using different datasets and data conditions, few studies have systematically compared them within a unified framework, particularly for plasma biomarkers such as p-tau181 and NfL. Such comparisons have been conducted in other domains. For instance, a 2025 study compared modelling approaches for mortality risk prediction in peritoneal dialysis patients, evaluating Cox proportional hazards models using baseline, time-averaged, and time-dependent biomarker values, as well as joint models (Goksuluk et al., 2025, pp. 1, 4-5). Overall, joint models demonstrated superior predictive performance, with multivariate joint models performing slightly better than separate joint models fitted for each biomarker (Goksuluk et al., 2025, p. 8). Given these findings, an additional question is whether models based on p-tau181, NfL, or their combination provide the best prediction of progression to AD dementia.

1.3 Research Aims

This study investigates the prediction of progression to AD-dementia using longitudinal blood biomarker data, comparing traditional and advanced statistical modelling approaches, as well as different biomarker specifications. More specifically, the research questions are:

1. What provides the best prediction of progression to Alzheimer’s disease-related dementia: accounting for baseline biomarker values only, longitudinal biomarker trajectories, or longitudinal biomarker trajectories together with latent class structure?

2. Across these modelling approaches, which biomarker strategy (NfL, p-tau181, or a combination of the biomarkers) provides the best prediction of progression to Alzheimer’s disease-related dementia?

2 Methodology

To address the research questions, a series of statistical models will be implemented. In all models, time to progression to AD-dementia, defined by the first recorded diagnosis of dementia, is the survival outcome. Given the relatively recent focus on blood-based biomarkers in AD research, their inclusion was considered timely and relevant. Blood-based p-tau181 and NfL were selected due to their established relevance for predicting dementia and AD progression, as well as the availability of rich longitudinal data with repeated measurements and overlapping follow-up information. In addition, they reflect two distinct pathological processes, making them suitable for comparative analyses of disease progression.

First, Cox PH models will be used to assess associations between baseline biomarker levels and progression risk. In these models, p-tau181 and NfL will be included either separately or jointly to allow comparison of their individual and combined effects. Second, JMs will be used to investigate how longitudinal biomarker trajectories are associated with the survival outcome. Initially, univariate JMs will be fitted for each biomarker separately. After that, multivariate JMs including both p-tau181 and NfL simultaneously will be applied to account for their combined and potentially correlated effects on disease progression. Finally, JLCMMs will be used to identify subgroups with distinct longitudinal biomarker trajectories and associated progression risks. As with the JM approach, both univariate and multivariate models will be evaluated.

All models will additionally be adjusted for baseline age, sex, and APOE- ϵ 4 status to improve predictive accuracy. In the JMs, these covariates will be included in both submodels, as they are assumed to have direct effects on both the longitudinal trajectory and the time-to-event process, ensuring coherent and well-specified predictions across the joint model components. The significance level will be set at 0.05 for all analyses.

Subsequently, the different modelling approaches will be compared based on their predictive performance. The same evaluation framework will also be used to compare different biomarker specifications, including p-tau181, NfL, and a combination of the two biomarkers. For prognostic models predicting progression from MCI to AD-dementia, predictions are most commonly made 2-4 years from baseline (Xia et al., 2025, p. 213). This reflects typical disease progression, where around one-third of individuals with MCI due to AD develop dementia within five years (Alzheimer’s Association, n.d.). Based on this, a landmark time of 1 year and prediction horizons of 2-6 years will be used to capture short- to mid-term progression. To ensure an unbiased assessment of model performance and generalisability, in-sample data will be used for model development with three quarters of the sample population, whilst out-of-sample data will be reserved for model evaluation with the remaining quarter.

The modelling approaches were chosen based on their methodological relevance and their ability to capture different aspects of the data, providing an overview of different approaches for modelling longitudinal biomarker data and survival outcomes. They will all be applied to data from the ADNI dataset, in accordance with its data use and access policies (ADNI, n.d.-b).

3 Statistical Theory

3.1 Survival Analysis

All of the modeling approaches considered in this study are grounded in the survival analysis framework. The primary aim is to predict the time until the occurrence of a pre-specified event of interest from some baseline time, usually the beginning of follow-up (Kleinbaum & Klein, 2012, p. 4). In this study, the event is defined as the first dementia diagnosis, consistent with previous dementia survival studies (Park et al., 2019, p. 4). Diagnosis status, ptau-181 and NFL levels are all available at baseline.

Additionally, individuals with AD-dementia are either considered as right censored, meaning that they do not reach the event during the follow-up period, or as interval censored, meaning that the exact date of the event is unknown (Kleinbaum & Klein, 2012, pp. 7, 286). Due to diagnoses exams being made either every half year or yearly, depending on the individuals cognition status and what stage of their protocol they are in, the exact date of transition to dementia can only be approximated. Therefore, participants in this study are considered as interval censored.

3.2 Cox Proportional Hazards Model

To quantify the association between biomarkers and disease progression, relative hazards models are commonly used. These models compare the hazard of progression between individuals with different covariate profiles, and assume that their hazards are proportional.

Given that T denotes either the time-to-event or the censoring time, the survival function models the probability of an individual remaining event-free up until time t , such that:

$$S(t) = Pr(T > t)$$

where the corresponding hazard function is given as the probability that the event occurs in the vicinity of time t divided by the probability of being event-free at time t :

$$h(t) = \frac{f(t)}{S(t)} \quad \text{where} \quad f(t) = \frac{d}{dt} [1 - S(t)]$$

and $1 - S(t)$ is the cumulative risk function, $F(t)$ (Moore, 2016, pp. 11-14). The Lehmann alternative for comparing survival functions assumes that two survival functions, $S_0(t)$ and $S(t)$, are related such that:

$$S(t) = [S_0(t)]^\psi$$

where ψ is the proportional hazards constant (Moore, 2016, pp. 43-44). This relationship can be re-expressed in terms of their hazard functions to give:

$$h(t) = \psi h_0(t)$$

which is the proportional hazards assumption (Moore, p. 55). Further, covariate information is linked to the proportional hazards constant exponentially, yielding the following model:

$$h(t) = h_0(t) \exp \{ \mathbf{x}^T \beta \}$$

where $\mathbf{x}$ is a vector of covariates and β is a vector of model parameters. In this study, the covariates of interest in the hazard model are age, gender, APOE- $\epsilon 4$ status and baseline values of p-tau181 and NfL. The latter two covariates may also be included simultaneously, resulting in a multivariable Cox PH model.

Relatedly, multicollinearity among the covariates should also be assessed. This occurs when at least two covariates in the regression model are highly correlated, potentially resulting in unstable estimates and misleading interpretations of the regression coefficients (Vatcheva et al., 2016, p. 2).

3.2.1 Model Assumptions

A key assumption of the Cox PH model is the assumed proportionality between hazard functions for different values of the specified covariates. This assumption is crucial for the construction of the partial likelihood, which allows the survival distribution to remain unspecified, since it allows the baseline hazard function to be dropped from the partial likelihood. At the same time, minor violations of this assumption are generally unlikely to have substantial effects on inference and parameter estimation (Moore, p. 94).

3.2.2 Concordance Index

The concordance index (C index) is an evaluation measure for the predictive performance of survival models. All pairs of individuals in the sample population are selected, and the C index evaluates the proportion of pairs who are concordant. Pairs are said to be concordant if the individual with a higher cumulative risk, given by $F(t)$, has the shorter time-to-event (Park et al., 2021, p. 1701). The C index will be considered as an initial evaluation metric for the different Cox PH models estimated in this study.

3.3 Joint Modelling of Longitudinal and Survival Data

In the joint model of longitudinal and survival data, the hazard function at time t is directly dependent on the value of the biomarker level at time t . In AD research, biomarkers such as p-tau181 and NfL are measured repeatedly over time, and the relationship between their trajectories and disease progression analysed.

The longitudinal process - modeled by a linear mixed-effects model - and the survival process - modeled by a proportional hazards model - are linked by a specified association structure. This structure can link the survival process to the current value of the biomarker estimate only, or include the current rate of change of the biomarker trajectory. For longitudinal data where measurements are repeated over the span of months, the current rate of change is better described as the difference between biomarker measurements. For instance, if both the current value and current rate of change are specified in the proportional hazards model, this becomes:

$$h(t|M_i, \mathbf{v}_i) = h_0(t) \exp \{ \gamma^T \mathbf{w}_i + \alpha_1 m_i(t) + \alpha_2 m'_i(t) \} \quad m'_i(t) = m_i(t) - m_i(t-1)$$

where α_1 and α_2 link the current value of the biomarker, $m_i(t)$, and its current rate of change, $m'_i(t)$, respectively to the hazard function. Further, γ is a vector of log-hazard ratios and $\mathbf{w}_i$ is a vector of time-independent covariates for the i^{th} individual.

This univariate framework is applied to the longitudinal p-tau181 and NfL measurements separately to model their trajectories jointly with survival outcomes. Multivariate joint models extend the standard joint modeling framework by allowing two or more longitudinal biomarkers to be modeled jointly while being linked to a shared survival outcome describing progression to AD or dementia. Therefore, both p-tau181 and NfL are considered simultaneously, as they capture complementary aspects of neurodegeneration and Alzheimer-related pathology. Ignoring their correlation may lead to biased estimation (Xu, 2021, p. 6)

3.3.1 Linear Mixed-Effects Submodel

The longitudinal component of the joint model is specified by the linear mixed-effects (LME) model (**LAIRD & WAIRE**), which captures both population-level trends and individual variability in biomarker trajectories.

Longitudinal data consists of repeated measurements from the same individuals, whereby the assumption of independence between observations is illogical. In this analysis, the levels of p-tau181 and NfL for an individual at a specific time point, t , are correlated with the measurements made on prior and subsequent occasions. Therefore, the responses from an individual are expected to be more highly correlated than those between individuals. The modelling of this type of data which can be considered grouped with non-independence between observations, is handled well by LME models.

Following the notation of Rizopoulos (2012, p. 54), the biomarker process for p-tau181 and NfL can be considered such that:

$$y_i(t) = m_i(t) + \epsilon_i(t) \quad \epsilon_i(t) \sim N(0, \sigma_\epsilon^2)$$

$$m_i(t) = \mathbf{X}_i^T \beta + \mathbf{Z}_i^T \mathbf{b}_i \quad \mathbf{b}_i(t) \sim N(0, \sigma_b^2)$$

where $y_i(t)$ is the observed biomarker measurement at time t for individual i . Assuming that the true biomarker value is not observed due to measurement and other errors, $m_i(t)$ then is the trajectory function which represents the true unobserved value of the biomarker at time t for patient i . The fixed effects are described by $\mathbf{X}_i^T(t)\beta$ and the random effects by $\mathbf{Z}_i^T \mathbf{b}_i$, where β are population-specific parameters and $\mathbf{b}_i$ are subject-specific parameters (Verbeke & Geert Molenberghs, 2000, p. 23). These subject-specific parameters handle the dependence between repeated measurements. Combining the true biomarker trajectory function with the error term gives:

$$y_i(t) = \mathbf{X}_i^T \beta + \mathbf{Z}_i^T \mathbf{b}_i + \epsilon_i(t) \quad \mathbf{b}_i(t) \sim N(0, \sigma_b^2), \quad \epsilon_i(t) \sim N(0, \sigma_\epsilon^2)$$

3.3.2 Survival Submodel

Assuming the survival submodel is defined as a Cox PH model, the longitudinal biomarker process is incorporated with the survival process, given a current value only association structure, such that:

$$h(t|M_i, \mathbf{v}_i) = h_0(t) \exp\{\gamma^T \mathbf{w}_i + \alpha m_i(t)\} \quad M_i = \{m_i, 0 \leq s \leq t\}$$

where M_i represents the true unobserved biomarker profile up to time t .

3.3.3 Model Assumptions

A core assumption of the joint model for longitudinal and survival data is full conditional independence between the longitudinal outcome and the survival outcome. All dependence is assumed to be explained by the random effects, resulting in independence of the longitudinal and survival outcomes and the repeated biomarker measurements are independent of each other. (Rizopoulos, 2012, pp. 62-63)

3.3.4 Estimation: Frequentist vs Bayesian Approach

The primary estimation approach that has been proposed for joint models is (semiparametric) maximum likelihood (Rizopoulos, 2012, p. 62). However, a key difficulty in joint modelling is that the estimation process can be very computationally demanding and may sometimes be infeasible (Wu et al., 2008).

A Bayesian alternative treats parameters as random variables and combines prior distributions with observed data to obtain posterior distributions, in contrast to the frequentist view where parameters are fixed but unknown (Carlin & Louis, 2008). This approach can simplify inference in complex models, and with weakly informative priors, it often yields results similar to maximum likelihood estimation (Brown & Ibrahim., 2003, p. 222).

Using the R package *JMbayes2*, joint models can be fitted via Markov chain Monte Carlo (MCMC) algorithms (Rizopoulos, 2018). MCMC generates dependent samples from the posterior distribution using a Markov chain, where each step depends only on the current state. As the chain evolves, its distribution converges to its stationary distribution, allowing the generated samples to be treated as representative of the posterior distribution and used for inference via Monte Carlo integration (Hamra et al., 2013, p. 628).

In *JMbayes2*, weakly informative ones are used, allowing stable estimation while letting the data primarily drive inference. The framework also accommodates multivariate settings, by introducing additional association parameters and regularising priors to handle higher-dimensional parameter structures (Rizopoulos, 2018).

3.3.5 The Gelman-Rubin Statistic

Within Bayesian joint modeling, convergence of the Markov chains can be assessed by examining whether the parameters have stabilised over iterations, for example through trace plots. Additionally, it can be assessed by running multiple chains with different starting values and comparing within-chain and between-chain variability, which is formalized by the Gelman–Rubin statistic (R) (Peng, 2022).

Consider J Markov chains, each with samples $x_1^{(j)}, x_2^{(j)}, \dots$ from the j -th chain. For each chain, remove the first D samples as “burn-in” and retain the following L samples: $x_D^{(j)}, x_{D+1}^{(j)}, \dots, x_{D+L-1}^{(j)}$. One common choice is to set $D = L$.

Then calculate the chain mean for each chain,

$$\bar{x}_j = \frac{1}{L} \sum_{t=1}^L x_t^{(j)}$$

the overall mean across all chains (the grand mean),

$$\bar{x}_\cdot = \frac{1}{J} \sum_{j=1}^J \bar{x}_j$$

the between-chain variance,

$$B = \frac{L}{J-1} \sum_{j=1}^J (\bar{x}_j - \bar{x}_\cdot)^2$$

the within-chain variance for each chain,

$$s_j^2 = \frac{1}{L-1} \sum_{t=1}^L (x_t^{(j)} - \bar{x}_j)^2$$

and finally, the average within-chain variance,

$$W = \frac{1}{J} \sum_{j=1}^J s_j^2$$

Subsequently, the Gelman-Rubin statistic is:

$$R = \frac{\frac{L-1}{L}W + \frac{1}{L}B}{W}$$

As the number of iterations L increases and the between-chain variation B decreases, R approaches 1. Thus, chains are typically run until R is near this number. Being a unit-free ratio, R provides a convenient summary of convergence without requiring specification of a particular parameter, making it a practical tool for monitoring MCMC chains before any inference is made (Peng, 2022).

3.4 Joint Latent Class Mixed Model

JLCMMs address suspected disease heterogeneity by assuming that the sample population consists of a finite number of unobserved subgroups (latent classes), each characterized by distinct longitudinal trajectories and survival risks (Kyheng et al., 2025, p. 2). Individuals within latent classes are assumed to share longitudinal biomarker trajectories and survival risks (Proust-Lima et al., 2018, p. 5) and the classes are not described by the observed covariates (Rizopoulos, 2012, p. 122).

In this analysis, univariate JLCMMs will be fit on the longitudinal p-tau181 and NfL measurements separately to assess whether any latent classes can be distinguished, and to ascertain any differences in predictive performance of these biomarkers.

DELETE IF WE DONT USE MULTI JLCMM: Multivariate JLCMM allowing multiple longitudinal biomarkers to be modeled simultaneously within each latent class, while linking their joint evolution to a class-specific risk.

, and then by compared to the multivariate model for both biomarkers,

3.4.1 Class Submodel

The JLCMM consists of three submodels; a multinomial logistic submodel, a linear mixed-effects submodel and a proportional hazards submodel. Multinomial logistic regression models classes such that:

$$\pi_{i,g} = Pr(c_i = g | \mathbf{X}_i) = \frac{e^{\mathbf{X}_i \beta_g}}{\sum_{l=1}^G e^{\mathbf{X}_i \beta_l}}$$

where $\mathbf{X}_i$ is the set of covaritates for the i^{th} individual and β_g is the set of coefficients for the g^{th} group. Given there are no prior assumptions about the latent classes, no covariates are specified in the multinomial logistic regression model (Kyheng et al., 2025, p. 2). The class submodel then simplifies to:

$$\pi_{i,g} = Pr(c_i = g) = \frac{e^{\beta_{0,g}}}{\sum_{l=1}^G e^{\beta_{0,l}}}$$

where $\pi_{i,g}$ is the probability that the i^{th} individual belongs to latent class g . Linking the class submodel to the longitudinal submodel and the survival submodel, assuming only a random intercept, results in the following:

$$h(t|c_i = g) = h_{0,g}(t) \exp \{ \gamma^T \mathbf{w}_i + \alpha_1 m_i(t) \}$$

3.4.2 Model Assumptions

In JLCMM, the biomarker and the time-to-event are considered to be conditionally independent given the latent classes, where all correlation between repeated measures of the biomarker is handled by the random effects in the longitudinal model (Proust-Lima et al., 2018, p. 122). This is in contrast to regular joint models, where the biomarker and time-to-event are assumed to be conditionally independent given the random effects (Proust-Lima et al., p. 62-63). The conditional independence assumptions for the JLCMM are as follows;

1. Correlations between observations of the longitudinal biomarker are modeled by the random effects, b_i , in the longitudinal model.
2. The relationship between biomarker trajectory and the time-to-diagnosis is captured completely by the latent class indicator, c_i .

POSSIBLY UNNECESSARY: There are various methods for assessing the conditional independence assumption, but simulations have demonstrated the superior performance of the score-test statistic in evaluating residual independence between the longitudinal and survival processes, even against two misspecified alternative hypotheses (Jacqmin-Gadda et al., 2010, p. 17).

The score test assumes the null hypothesis of conditional independence, that is, all correlation between the biomarker trajectory and time-to-event is captured entirely by group membership. Given the class membership is known, there is assumed to be no remaining dependence between the biomarker response and the time-to-event response. Assuming a model with a current value association structure, the association parameter, α_1 , is assumed to be zero under the null hypothesis (Jacqmin-Gadda et al., p. 13), such that:

$$H_0 : \alpha_1 = 0 \quad H_A : \alpha_1 \neq 0$$

Under the null hypothesis, the score test statistic, U , is χ^2 distributed with degrees of freedom equal to the number of shared-random effects in the model (Proust-Lima et al., p. 8). A significant test statistic supports the alternative hypothesis that there is residual dependence between the biomarker and time-to-event responses.

3.4.3 Estimation

The log-likelihood for the JLCMM is much more tractable than for the regular joint model (Rizopoulos, 2012, p. 123), making estimation less computationally demanding. Log-likelihood estimation for JLCMM involves summing over the latent classes (Proust-Lima et al., 2014, p. 5), which is computationally advantageous when compared with the estimation of joint models which require mathematical approximation through, for example, the previously defined MCMC algorithm, due to integration over the random effects. For JLCMM, the log-likelihood

is maximised using a **Marquardt algorithm**. In model estimation it should be considered that there may be several local maxima, meaning the algorithm should be run several times using different starting values (Proust-Lima et al., 2014, p. 5).

In the *lcmm* package developed by Proust-Lima et al. for estimating JLCMMs, a Weibull baseline hazard model is estimated by default and it is not possible to estimate a Cox PH model, in order to stay in the full-likelihood framework (Proust-Lima et al., 2014, p. 5). Therefore, it is necessary to assess whether the hazard is best modeled by the Weibull or by a flexible baseline hazard model.

3.4.4 Class Enumeration

Determining the number of latent classes (class enumeration) is one of the challenges of model selection for JLCMMs. Likelihood ratio tests cannot be performed for nested latent class models with differing number of classes, as regularity conditions which ensure a χ^2 distributed test statistic do not hold (Celeux & Soromenho, 1996, p. 196). Therefore, other metrics must be utilised for this purpose.

One aspect of determining the optimal number of classes is minimization of the Bayesian Information Criterion (BIC), which is the preferred criterion for mixture models (Proust-Lima et al., 2014, p. 7). Classes must also be adequately discriminated, which is assessed by analysis of posterior class membership probabilities, given by (Proust-Lima et al., 2014, p. 7):

$$\hat{\pi}_{i,g} = P(c_i = g | Y_i, (T_i^*, E_i); \hat{\theta}_G) = \frac{\hat{\pi}_{i,g} f(Y_i | c_i = g; \hat{\theta}_G) \lambda_i(T_i | c_i = g; \hat{\theta}_G) S_i(T_i | c_i = g; \hat{\theta}_G)}{\sum_{l=1}^G \hat{\pi}_{i,l} f(Y_i | c_i = l; \hat{\theta}_G) \lambda_i(T_i | c_i = l; \hat{\theta}_G) S_i(T_i | c_i = l; \hat{\theta}_G)}$$

Reporting the proportion of individuals who have a posterior probability over a certain threshold (Proust-Lima et al., 2014, p. 7) - typically 0.7, 0.8 or 0.9 - illustrates the ability of the model to discriminate classes from one another. Mean posterior probabilities for each class can also be reported, with $\mu(\hat{\pi}_g)$ values close to 0.5 showing poor class discrimination. Class sizes are also important, as classes that are made up of less than 5% of the total sample population may be driven by outliers and therefore not be clinically meaningful.

Entropy

One possible measure of the discriminative power of a latent class structure is relative entropy (Ramaswamy et al., 1993 in Morgan et al., 2016, p. 150), which takes values between 0 and 1 and is calculated as follows:

$$EN(G) = 1 - \frac{\sum_{i=1}^N \sum_{g=1}^G \hat{\pi}_{i,g} \log(\hat{\pi}_{i,g})}{N \log(G)}$$

where G is the number of latent classes specified by the model and $\hat{\pi}_{i,g}$ is the posterior probability that the i^{th} individual belongs to class g . Relative entropy values closer to 1 demonstrate good discriminative ability of the model, where, for one class latent models, $EN(1) = 1$.
***CONCEPTUALLY, WHAT DOES THIS MEASURE???**

Validation

The chosen classes should be clinically meaningful.

3.5 Evaluation of Predictive Performance

Information criteria such as the Akaike Information Criterion (AIC) are often used to evaluate predictive performance in joint models combining longitudinal and survival data. However, when the main focus is the survival outcome and how well a longitudinal biomarker predicts it, performance is typically assessed using two main approaches: calibration, which measures how closely predicted probabilities agree with observed outcomes, and discrimination, which evaluates how well the model distinguishes between individuals who experience the event earlier versus later (Rizopoulos, 2012, p. 194). Time-dependent Area Under the Curve (AUC) is an example of a discrimination measure, while the time-dependent Brier score is a calibration-oriented measure that also reflects overall prediction error over time.

3.5.1 Time-dependent AUC

Receiver Operating Characteristic (ROC) analysis is commonly used to evaluate the ability of a marker or model to discriminate between binary outcomes, such as diseased and non-diseased individuals (Zou et al., 2012, p. 3). However, the traditional ROC framework assumes that both disease status and marker values are fixed over time (Kamarudin et al., 2017, p. 1).

In survival settings, this assumption is often violated, as event status changes during follow-up and marker values may vary over time. To account for this, time-dependent ROC curves have been developed, incorporating the timing of events and censoring, and providing a more appropriate framework for assessing prognostic accuracy (Kamarudin et al., 2017, p. 1).

When a time-dependent framework is adopted, the disease status is evaluated at each time point, meaning that sensitivity and specificity may change over the course of the study. Let T_i denote the time of disease onset for individual i , and let X_i represent a marker value (typically measured at baseline) for individual i , where $i = 1, \dots, n$ (Kamarudin et al., 2017, p. 2).

QUESTIONS:

1. When is Z_i used after defining it?
2. What is the difference between δ_i and $D_i(t)$? Does δ_i refer to whether the event happens at all, whereas $D_i(t)$ refers to whether the event has occurred by time t ? If they are different, where is δ_i referenced after being defined?

The observed time is defined as $Z_i = \min(T_i, C_i)$, where C_i is the censoring time. In addition, let δ_i be an indicator variable that takes the value 1 if the event (disease) occurs and 0 otherwise. Furthermore, let $D_i(t)$ denote the disease status at time t , where $D_i(t) = 1$ indicates presence of disease at time t , and $D_i(t) = 0$ otherwise (Kamarudin et al., 2017, p. 2).

In what follows, X is referred to as a marker, although it may also represent a risk score derived from a regression model or another predictive model, including previously published scores (Kamarudin et al., 2017, p. 2).

For a given threshold c , the time-dependent sensitivity and specificity are defined as:

$$Se(c, t) = P(X_i > c \mid D_i(t) = 1)$$

$$Sp(c, t) = P(X_i \leq c \mid D_i(t) = 0)$$

Based on the definitions above, the corresponding receiver operating characteristic (ROC) curve at a given time t , denoted by $ROC(t)$, can be constructed by plotting the time-dependent sensitivity $Se(c, t)$ against $1 - Sp(c, t)$ across all possible thresholds c (Kamarudin et al., 2017, p. 2).

The time-dependent area under the curve (AUC) is then defined as:

$$AUC(t) = \int_{-\infty}^{\infty} Se(c, t) d[1 - Sp(c, t)]$$

An AUC value of 1 means perfect ability to distinguish between patients with and without the condition, while an AUC of 0.5 means no discrimination at all, meaning the marker performs no better than random guessing such as flipping a coin (Rizopoulos, 2012, p. 196).

3.5.2 Time-dependent Brier score

The Brier score can be thought of as the mean squared error of a prediction forecast. However, for survival data, the true event status cannot be known for censored individuals. Therefore, an inverse-probability-of-censoring weighting (IPCW) adjusted Brier score has been formulated to account for bias due to censoring, where those with earlier event times are over-represented, leading to underestimation of survival probabilities (Prince et al., 2025, p. 4). IPCW weights are calculated for each individual $i = 1, \dots, n_{sample}$ at each time point t , such that:

$$\omega_i(t) = \frac{\mathbb{I}\{T_i^* \leq t, \delta_i = 1\}}{G_i(T_i^* -)} + \frac{\mathbb{I}\{T_i^* > t\}}{G_i(t)}$$

where $G_i(t)$ is the censoring survival function, which quantifies the probability of surviving a censoring event for the i^{th} individual, T_i^* is a time point infinitesimally smaller than the observed time T_i^* , and $\mathbb{I}\{\cdot\}$ is an indicator function (Prince et al., 2025, pp. 4-5). The commonly formulated Brier score:

$$BS = \frac{1}{n_{sample}} \sum_{i=1}^{n_{sample}} (Y_i(t) - \widehat{S}_i(t))^2$$

where $\widehat{S}_i(t)$ is the predicted probability of an event for the i^{th} individual at time t and $Y_i(t)$ is their observed outcome at that time (Park et al., 2021, p. 1704), becomes then the IPCW-adjusted Brier score:

$$BS = \frac{1}{\tilde{n}(t)} \sum_{i=1}^{n_{sample}} \left(\frac{\widehat{S}_i(t)^2 \mathbb{I}\{T_i^* \leq t, \delta_i = 1\}}{G_i(T_i^*-)} + \frac{(1 - \widehat{S}_i(t))^2 \mathbb{I}\{T_i^* > t\}}{G_i(t)} \right)$$

where $\tilde{n}(t)$ is the sum of the IPC weight (Prince et al., 2025, p. 6). The Brier score takes values from 0 to 1, where a score closer to 0 is given to a model with better predictive ability.

4 Data

The data used in this analysis comes from the ADNI cohort under the ADNI1, ADNIGO and ADNI2 protocols. The variables of interest were all retrieved from the ADNIMERGE2 package, found in the ADNI database (ADNI, n.d.-b). In accordance with the ADNI leadership guidelines, which state that there were serious data quality concerns for individuals with a PTID in the format “381_S_10#” , checks were run to assure that no observations with these PTIDs were present in the data set (ADNI, 2026).

The inclusion criteria for this study require participants to have recorded diagnostic visits and biomarker measurements of both p-tau181 and NfL. The exclusion criteria was any individual with a baseline dementia diagnosis. This resulted in a sample size of 934 individuals with at least one biomarker measurement.

Moreover, the inclusion criteria also concerns the number of biomarker measurements per individual and is set to at least three measurements. However, sensitivity analyses will also be applied, where the analysis will only require individuals to have at least one biomarker. This is based on a common issue in modeling longitudinal biomarker data, which is whether to include or exclude individuals with sparse data in the analysis (Rizopoulos, 2012, pp. 27-28). While the previously illustrated studies included all participants regardless of the number of measurements per individual, a study examining different cognitive biomarkers to predict progression to AD included only those with at least three complete longitudinal cognitive assessments (Guo et al., 2026, p. 2). Other studies have restricted the inclusion criteria

further to those with at least ten longitudinal biomarker measurements in order to ensure the approximation did not lead to non-ignorable biases (Wang et al., 2024, pp. 591-592).

Participants with cross-sectional data cannot provide information about within-subject change over time, however they still contribute useful information about between-subject variability (Fitzmaurice et al., 2011, pp. 214-215). Complete data analyses risk loss in precision dependent on the pattern and distribution of missing data across subjects and time as well as the missing data mechanism. The main concern of missing data is the potential for introducing bias, resulting in misleading inferences. Consequently, the mechanism underlying missing data is a critical consideration in longitudinal analyses, as it determines whether exclusion of incomplete cases may distort results (Fitzmaurice et al., 2011, p. 491). The handling of individuals with sparse data reflects an ongoing methodological discussion regarding how longitudinal data should be handled in practice.

This reduced to 704 when restricting the sample to participants with three or more biomarker measurements.

All missing values were removed prior to the analysis to ensure a complete-case data set and avoid bias introduced by incomplete observations.

4.1 Variables

4.1.1 Diagnosis status

The variable of primary interest represents three diagnostic states - cognitively normal (CN), mild cognitive impairment (MCI) and dementia - and was retrieved from the DXSUM data set in ADNIMERGE2. Diagnostic status is assessed over several examination dates, with the number of diagnostic exams differing depending on the baseline diagnosis.

Since the study focuses on time to incident dementia, participants are classified as either non-demented or demented for the survival analyses, without distinguishing between CN and MCI states. Diagnostic records occurring more than 50 days prior to the first biomarker measurement were excluded to maintain temporal alignment between clinical and biomarker data.

When restricting the participants to those with three or more biomarker measurements, the sample size of the in-sample data set was 528, among which 128 experienced the event of dementia, corresponding to 24.24%, while 400 individuals were censored. In the out-of-sample, there were 176 individuals, among whom 49 experienced the event of dementia, corresponding to 27.84%, while 127 individuals were censored. In comparison, extending the sample population to those with at least one biomarker measurement, the in-sample data set consisted of 701 individuals, among whom 145 experienced the event of dementia, corresponding to

20.69%, while 556 individuals were censored. The out-of-sample data set included 233 individuals, among whom 53 experienced the event of dementia, corresponding to 22.75%, while 180 individuals were censored.

4.1.2 p-tau181 and NfL

The p-tau181 measurements were retrieved from the UGOPTAU181 data set, which was processed by the University of Gothenburg (ADNI, n.d.-b), while the NfL measurements were retrieved from the BLENNOWPLASMANFLLONG data set, processed by the Blennow Lab (ADNI, n.d.-b). The unit of measurement is not specified in the data set documentation for either p-tau181 or NfL.

Among participants with at least three biomarker measurements, p-tau181 values ranged from 0.4 to 238.3, with a median of 15.5. Similarly, NfL values ranged from 6.9 to 408.0, with a median of 36.5. The pronounced right-skewness for both biomarkers was the basis for log-transformation in order to better satisfy the assumption of normality (Appendix 8.1; Figure 6a & 6b). A similar right-skewed distribution was observed when including all participants with at least one biomarker measurement. Values larger than three standard deviations were analysed and it was determined that they should be retained in the analysis. It is also worth noting that the correlation between log-transformed p-tau181 and NfL was 0.34 (8.1.4), indicating a weak positive association. This suggests that the biomarkers share some common variation while largely capturing distinct biological processes. Overall, this supports the use of a multivariate framework, as it allows their dependence to be accounted for while evaluating both their individual and combined associations with disease progression.

Biomarker examination dates ranged from 1 February 2007 to 6 June 2016, while diagnostic assessment dates spanned 29 September 2005 to 7 November 2025. Diagnoses were therefore available several years prior to the first biomarker measurements. In this analysis, baseline was defined as the first biomarker examination date, ensuring alignment of biomarker follow-up with dementia onset.

52 participants had two measurements with different values recorded on the same examination date. To ensure consistency, the mean value of these measurements was selected as the primary approach. **Sensitivity analyses showed no substantial differences in model precision: did we do this yet?** Furthermore, discrepancies were observed between p-tau181 and NfL examination dates with the same visit code. Five cases exceeded 30 days, with differences of up to 231 days, indicating that these did not represent the same examination occasion. To address this, a common examination date was constructed using the midpoint between the two biomarker dates.

4.1.3 Age

Additionally, age is included as a covariate when predicting progression to dementia, as risk of dementia is known to increase with age (Alzheimer’s Society, n.d.). The variable was retrieved from the PTDEMOG data set and was initially recorded as year of birth for each participant (ADNI, n.d.-b) which was transformed into age at baseline. Baseline age ranged from 55 to 94 years, with a median age of 74 years.

4.1.4 Gender

Another covariate included when predicting progression to dementia is participants’ sex at birth, as recorded on their original birth certificate, due to reported differences in disease risk and progression between sexes. More precisely, women aged over 80 have a slightly higher risk of developing Alzheimer’s disease compared to men of the same age (Alzheimer’s Society, n.d.). This binary variable includes two categories: male and female (ADNI, n.d.-b). As for age, the variable was retrieved from the PTDEMOG dataset.

4.1.5 APOE4-status

Finally, APOE- ϵ 4 status is included as a covariate when predicting progression to dementia, due to its well-established role as a genetic risk factor for AD. The ϵ 4 allele of apolipoprotein E (APOE) increases the risk of AD, whereas the - ϵ 2 allele has been shown to have a protective effect (Wong et al.). The variable was retrieved from the APOERES dataset (ADNI, n.d.-b). Initially coded numerically based on the number of ϵ 4 alleles, the variable was recoded into three ordinal categories: 0 alleles (non-carriers), 1 allele (heterozygous carriers; 2/4 or 3/4), and 2 alleles (homozygous carriers; 4/4).

Among the participants with at least three biomarker measurements, 83 participants carried no APOE- ϵ 4 alleles, 578 carried one allele, and 43 carried two alleles. When looking at the participants with at least one biomarker measurement, 109 of these participants carried no APOE- ϵ 4 alleles, 772 carried one allele, and 53 carried two alleles.

5 Results

5.1 Model Fit Overview

The initial analysis was run on the restricted cohort of individuals with three or more longitudinal p-tau181 and NfL measurements. A subsequent sensitivity analysis was run on the extended cohort including all individuals with at least one p-tau181 and NfL measurement in order to verify whether the results are stable when including individuals with sparse longitudinal data.

5.1.1 Cox PH Models

First, Cox PH models including baseline values of p-tau181, NfL, or both were estimated. Additional covariates were gender, age at baseline and APOE genotype. Initially, fitting the model with all levels of genotype - which denotes the specific APOE allele combination - resulted in non-significant coefficients for all levels. Therefore, genotype was collapsed into three levels denoting the presence of the APOE- $\epsilon 4$ allele, taking the following values: 0 alleles (non-carriers), 1 allele (heterozygous carriers; $\epsilon 2/\epsilon 4$ or $\epsilon 3/\epsilon 4$), and 2 alleles (homozygous carriers; $\epsilon 4/\epsilon 4$).

Prior to interpreting the predictive performance and the estimated hazard ratios of the models, the proportional hazards assumption was assessed for all covariates to verify that the effects remained constant over time. Considering the Cox PH model is defined as the survival sub-model in the subsequent JMs, it is critical this assumption is met in order to reliably evaluate the JMs. The resulting p -values are presented in Table 14 (Appendix 8.2).

At a 5% significance level, most covariates satisfied the proportional hazards assumption across all models, as indicated by non-significant p -values for sex, APOE- $\epsilon 4$ status, and the biomarker variables. However, age at baseline consistently showed significant p -values, indicating a violation of the assumption. The same pattern was observed for the GLOBAL test, which jointly evaluates the proportional hazards assumption across all covariates and is sensitive to covariate-specific departures from the assumption; it also yielded significant or borderline significant p -values.

Given this violation, age at baseline was excluded from the final Cox PH models. The refitted models yielded the p -values presented in Table 1.

Table 1: p -values for the proportional hazards assumption tests after exclusion of age at baseline in the Cox PH models

variable	Standard	p-tau181 baseline	NfL baseline	Both baseline
PTGENDER	0.864	0.875	0.849	0.907
APOE4_status	0.178	0.186	0.134	0.158
GLOBAL	0.324	0.448	0.163	0.266
baseline_value_ptau	NA	0.692	NA	0.623
baseline_value_nfl	NA	NA	0.134	0.108

After excluding age at baseline, the GLOBAL test yielded p -values above the 5% significance level, no longer indicating any violation of the proportional hazards assumption. All other covariates also remained non-significant, suggesting that they jointly do not exhibit systematic time-dependent effects in any of the refitted models.

Furthermore, an initial comparison of model performance by assessment of the concordance indices for all cox models was performed. The results presented in Table 2 show that the

concordance index evaluates all three models similarly, with results indicating adequate discrimination.

Table 2: Concordance indices for the fitted Cox PH models with baseline biomarker values

Model	C-Index
Cox ptau-181	0.679
Cox NfL	0.673
Cox ptau-181 & NfL	0.681

5.1.2 Joint Models

Next, JMs that incorporated the longitudinal trajectories of p-tau181 and NfL with the time-to-diagnosis data were estimated. For the longitudinal component, an LME model was estimated separately for p-tau181 and NfL.

Plotting the residuals against the fitted values for the LME model with p-tau181 showed no clear evidence of heteroskedasticity, as shown in Appendix 8.3; Figure 8a. The slight deviations in the tails of the Q-Q plot were also not considered to be of concern (Appendix 8.3; Figure 8b). A similar pattern was observed for the LME model of NfL, where the plot of residuals against fitted values demonstrated even less evidence of heteroskedasticity compared to the corresponding plot for p-tau181 (Appendix 8.3; Figure 9a). The Q-Q plot for NfL showed similar results to those observed for p-tau181 (Appendix 8.3; Figure 9b). Generalised LME models with different link functions were also fit and their residuals analysed, however none of these models demonstrated a better fit than the LME models with log-transformed biomarker measurements. Consequently, they were considered to provide an adequate fit to the data and as suitable for use within the joint modeling framework.

The choice of random effects structure for the JMs was limited due to the *JMBayes2* package currently only supporting those including a random slope. Therefore, all LME models were fit with a random intercept and slope. However, models with a simple structure with only a random intercept and an extended structure including a random slope were fit using maximum likelihood estimation in order to perform likelihood ratio tests. The results show that the extended structure was a better fit for NfL, but a simpler structure would have been preferable for p-tau181 (Appendix 8.2; Table 13).

Thus, JMs with a random intercept and slope were estimated for longitudinal measurements of p-tau181, NfL, or both simultaneously. For each model, three MCMC chains were used, with 11,000 iterations and 1,000 burn-in iterations for the univariate JMs, and 20,000 iterations with 2,000 burn-in iterations for the multivariate JM. To ensure convergence of the JMs, trace plots of the α -parameters, which represent the associations between the longitudinal and survival submodels, were examined for each model. The evaluation started with the JM for p-tau181, presented in Figure 1.

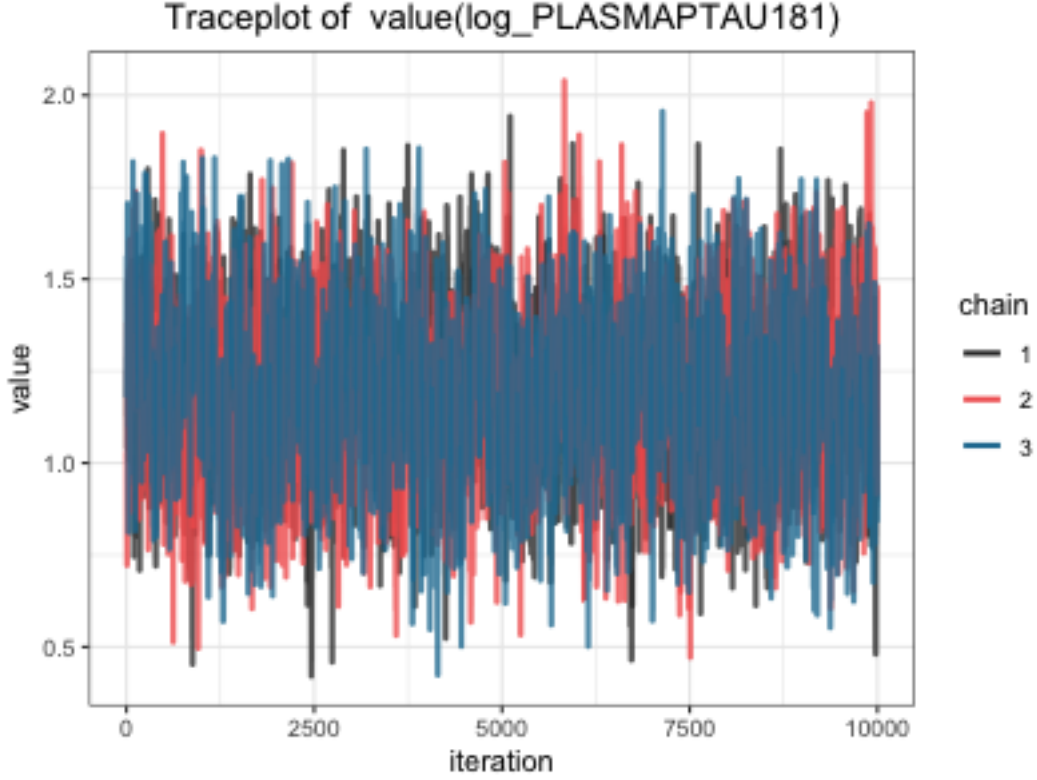


Figure 1: Traceplot of log-transformed p-tau181

The trace plot shows that the three MCMC chains fluctuate around a stable mean value throughout the iterations, without any clear trend or drift. This suggests that the chains have reached a stable region. In addition, the chains overlap well and mix with each other, indicating that they are sampling from the same posterior distribution. Although some random variation is present, this is expected in MCMC sampling. Overall, the plot supports good convergence.

A similar pattern was observed for the JM including NfL, as shown in Appendix 8.4; Figure 10a. The chains likewise fluctuate around a stable mean value with good mixing, indicating convergence of the α -parameters comparable to the JM including p-tau181.

Finally, the multivariate JM also exhibited a similar overall behaviour (Appendix 8.4; Figure 10b-c). Compared to the single-biomarker models, however, the trace plot for the multivariate JM shows a slightly wider range of α -values, indicating a somewhat larger variability in the α -parameters. This suggests increased uncertainty in the parameter estimates, which is expected given the higher complexity of the model. Despite this, the chains still fluctuate around a stable mean value with no visible trend or drift, and they continue to overlap and mix well. Thus, all three models appear to have converged adequately.

Additionally, convergence of the models was assessed by examining the $\widehat{R}$ values of the survival outcome and the longitudinal outcome in each joint model, with the survival $\widehat{R}$ values presented in Table 3.

Table 3: $\widehat{\mathbf{R}}$ values of the survival outcome in the joint models

Variable	p-tau181	NfL	Multivariate
PTGENDERMale	1.001	1.001	1.002
APOE4_status1 allele	1.003	1.002	1.003
APOE4_status2 alleles	1.002	1.002	1.002
value(log_PLASMAPTAU181)	1.003	NA	1.008
value(log_PLASMA_NFL)	NA	1.008	1.010

The $\widehat{R}$ values for the survival outcomes remained very close to 1, ranging from 1.000 to 1.007 across all JMs. This indicates good convergence of the MCMC chains for these components and confirms the convergence patterns observed in the α -parameter trace plots, with no evidence of convergence issues.

While Table 3 presents the $\widehat{R}$ values for the survival outcomes, the corresponding $\widehat{R}$ values for the longitudinal outcomes in each JM are shown in Table 4.

Table 4: $\widehat{\mathbf{R}}$ values of the longitudinal outcome in the joint models

Variable	p-tau181	NfL	Multivariate
(Intercept)	1.003	1.000	1.002
time_years	1.049	1.005	1.061
PTGENDERMale	1.000	1.000	1.000
APOE4_status1 allele	1.000	1.000	1.000
APOE4_status2 alleles	1.001	1.001	1.001
sigma	1.003	1.004	1.008

In a similar manner, the $\widehat{R}$ values for the longitudinal outcomes are generally close to 1 across all JMs, indicating good convergence for most parameters. The intercepts and covariate effects show values very close to 1, suggesting stable convergence. A slightly higher value can be observed for the time effect in the multivariate model, where $\widehat{R}$ reaches 1.09. This suggests somewhat slower convergence for this parameter in the multivariate model, likely reflecting the increased complexity of the longitudinal structure. Overall, when considered together with the trace plots and survival $\widehat{R}$ values, the results still support adequate convergence of the models.

5.1.3 Joint Latent Class Mixed Models

Finally, JLCMMs were estimated. The models were selected via comparison of BIC and entropy as well as assessment of posterior probabilities and classifications for models with up to four latent classes. An intercept only random effects structure was chosen due to minimised BIC for all models. A Weibull baseline hazard was chosen and checked against whether a flexible baseline hazard with the same number of parameters resulted in a lower BIC, which was the case for all models. The results of each model were verified using the *gridsearch()* function which allows for better exploration of the parameter space to ensure finding the global instead of a local maximum solution. This check confirmed that using random starts from the one class model was sufficient for finding the global maximum for all models.

The log-likelihood, BIC and entropy of the models using longitudinal p-tau181 measurements are presented below (Table 5). Log-likelihood and entropy increased with additional latent classes, however the BIC was minimised for the model with two latent classes.

Table 5: Log-likelihood, BIC and entropy for JLCMMs with longitudinal p-tau181 measurements

Candidate model	G	Log Likelihood	BIC	Entropy	Class1 (%)	Class2 (%)	Class3 (%)	Class4 (%)
Model1	1	-2019.695	4077.005	1.000	100.00	NA	NA	NA
Model2	2	-1989.499	4041.688	0.648	24.43	75.57	NA	NA
Model3	3	-1984.353	4056.474	0.661	3.79	71.21	25.00	NA
Model4	4	-1974.834	4062.511	0.714	22.92	3.03	67.42	6.63

Posterior classifications show the number of individuals assigned to each class, whereas the mean posterior probabilities show, for individuals in each class, the average probability of being assigned to each class. For example, values for *class1*, *prob1* close to 1 and *class2*, *prob1* close to 0 show good class discrimination.

For the two class model (Table 6), the class sizes are reasonable and the mean posterior probabilities show good class discrimination, meaning the model has high certainty in assigning individuals to their respective classes. Comparison with the three class model (Table 7) shows somewhat poorer class discrimination as well as a questionably small *class3*, with only 3.79% of individuals being assigned to this class.

Table 6: Posterior classification table and mean posterior probabilities for the two class JLCMM for longitudinal p-tau181 measurements

	Prob1	Prob2	Prob > 0.7 (%)	Prob > 0.8 (%)	Prob > 0.9 (%)
Class1	0.9671	0.0329	93.80	91.47	91.47
Class2	0.1204	0.8796	90.98	76.19	52.63

Table 7: Posterior classification table and mean posterior probabilities for the three class JLCMM for longitudinal p-tau181 measurements

	Prob1	Prob2	Prob3	Prob > 0.7 (%)	Prob > 0.8 (%)	Prob > 0.9 (%)
Class1	0.7429	0.2388	0.0183	65.00	45.00	15.00
Class2	0.0521	0.8242	0.1237	86.17	62.77	28.72
Class3	0.0072	0.0433	0.9494	93.18	87.88	84.09

The score test statistic for the conditional independence assumption was significant for the one class model (50.336, $p < 0.001$) but became non-significant (0.122, $p = 0.727$) upon modeling two latent classes, further evidence that the data supports two latent classes in the sample population. All model parameters were significant ($p < 0.001$) and are presented in Table 16 (Appendix 8.5). Therefore, the two class model was chosen for this analysis.

The log-likelihood, BIC and entropy provided in Table 8 for candidate models using longitudinal NFL data suggest that also in this instance the two class model provided the best fit for the data and had the highest classification certainty. Log-likelihood increased most between the one class and the two class model, whilst only a slight increase was exhibited between the two and three class models. BIC was minimised and entropy was maximised for the two class model.

Table 8: Log-likelihood, BIC and entropy for JLCMMs with longitudinal NFL measurements

Candidate model	G	Log Likelihood	BIC	Entropy	Class1 (%)	Class2 (%)	Class3 (%)
Model1	1	-1060.314	2158.243	1.000	100.00	NA	NA
Model2	2	-1032.385	2127.461	0.688	76.33	23.67	NA
Model3	3	-1031.169	2150.106	0.471	73.11	1.89	25

Class sizes were reasonable and good class discrimination was shown when assessing the average posterior probabilities (Table 9). The one class model had a significant score test statistic (33.409, $p < 0.001$) which became non-significant for the two class model (0.368, $p = 0.544$). The two class model was therefore chosen.

Table 9: Posterior classification table and mean posterior probabilities for the two class JLCMM for longitudinal NFL measurements

	Prob1	Prob2	Prob > 0.7 (%)	Prob > 0.8 (%)	Prob > 0.9 (%)
Class1	0.8982	0.1018	93.55	82.38	60.3
Class2	0.0270	0.9730	96.00	95.20	92.8

5.2 Model Comparison

In order to compare the different modelling approaches, predictive performance was evaluated using time-dependent AUC at yearly time points from a landmark time of one year over horizons of one to five years. Figure 2 provides the time-dependent AUC of the different models across all prediction horizons, with the biomarker fixed as p-tau181.

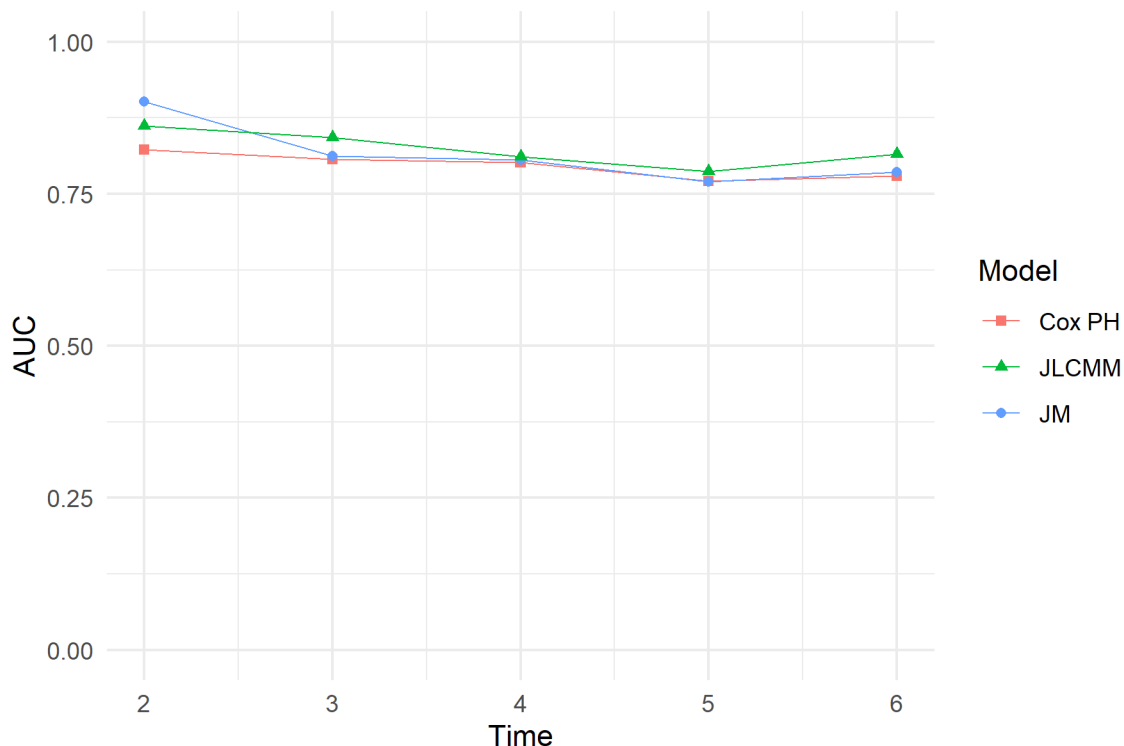


Figure 2: Time-Dependent AUC (p-tau181)

Among these models, the time-dependent AUC curves remain consistently high, with values exceeding 0.75 for all horizons, regardless of modelling approach. This indicates that the models demonstrate stable and relatively strong overall predictive performance. The values are slightly higher for earlier predictions, especially for the JM. Differences between the models decrease over time, resulting in increasingly similar curves, yet the AUC values suggest that the best predictions made at year six were from the JLCMM. The Cox PH model consistently yields the lowest AUC values.

Among the models incorporating NfL, a similar overall pattern is observed. As shown in Appendix 8.6; Figure 11a, the curves remain at comparable levels to those seen for the p-tau181 models, with time-dependent AUC values exceeding 0.75 across all horizons. For earlier predictions, where the values are slightly higher across models, the JM once again yields the

highest AUC values. These values then decline over time, becoming more similar to those of the other models. In addition, it shows higher a slightly higher value for the JLCMM at for the latest prediction horizon.

Appendix 8.6; Figure 11b presents the models incorporating both biomarkers simultaneously, demonstrating higher AUC values for early predictions made using the JM. These values gradually decline over time, yielding values more similar to those of the Cox PH model for later predictions. These results are also indicated by Table 10 which contains the AUC values for each modelling approach and at each horizon.

Table 10: Time-dependent AUC by model and prediction time point

Model	AUC_2y	AUC_3y	AUC_4y	AUC_5y	AUC_6y
Cox ptau-181	0.823	0.807	0.802	0.771	0.780
Cox NfL	0.846	0.787	0.817	0.792	0.797
Cox both	0.870	0.847	0.856	0.837	0.855
JM ptau-181	0.902	0.812	0.806	0.770	0.786
JM NfL	0.953	0.799	0.823	0.808	0.804
JM both	0.967	0.859	0.863	0.841	0.846
JLCMM ptau-181	0.862	0.843	0.811	0.787	0.815
JLCMM NfL	0.907	0.800	0.803	0.809	0.837

For instance, by showing that the JM among the p-tau181-based models has an AUC value of 0.902 for predictions made at year two, while the Cox PH model has an AUC value of 0.823 and the JLCMM has an AUC value of 0.862, Table 10 confirms the results previously observed in the figure. The differences between the models then decrease over time; for predictions at four years, the AUC values are 0.802 for the Cox PH model, 0.806 for the JM, and 0.811 for the JLCMM.

When evaluating predictive performance using time-dependent Brier scores, the same overall patterns are observed, with the key difference that lower values, and therefore lower curves, indicate better predictive performance. Figure 3 presents the time-dependent Brier score curves for the different modelling approaches, with the biomarker specification fixed to p-tau181.

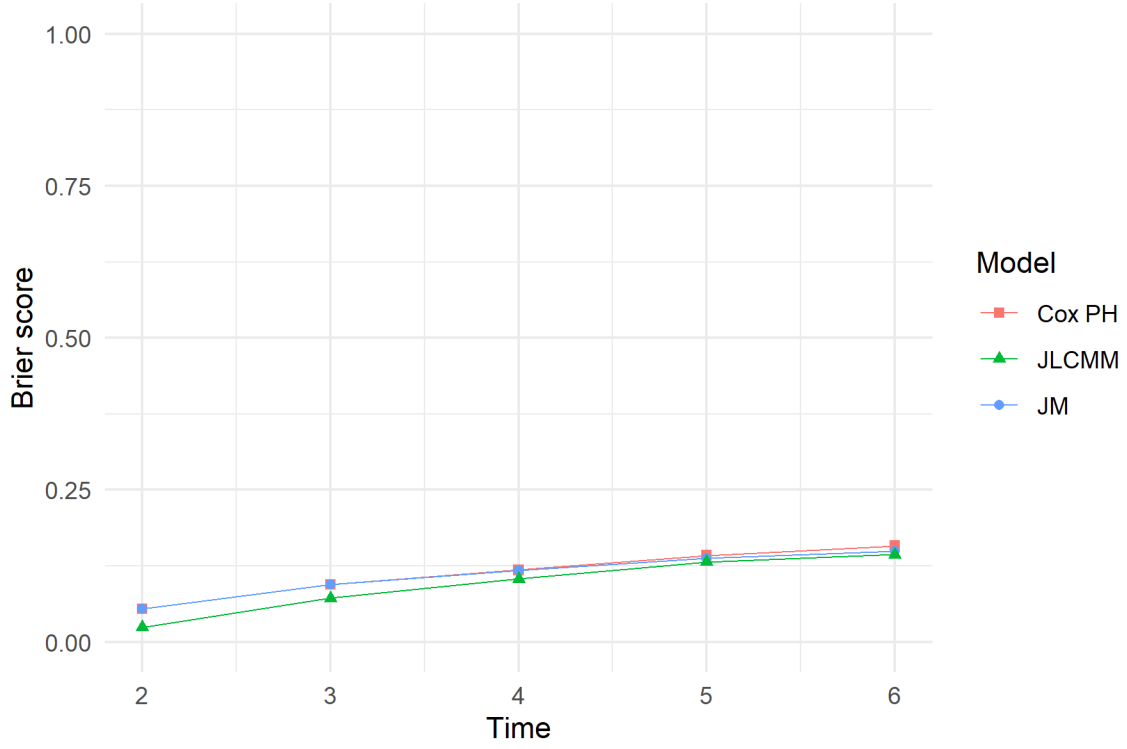


Figure 3: Time-dependent Brier scores (p-tau181)

Among the models involving p-tau181, the time-dependent Brier score curves remain low, with values ranging from approximately 0.05 for early predictions to 0.16 at later horizons, regardless of modelling approach. This suggests that the models demonstrate stable and relatively strong overall predictive performance. For earlier horizons, however, the JLCMM yields lower Brier score values compared with the other two models, indicating better predictive performance. The differences between models decrease for later predictions, as the curves get increasingly closer.

This pattern holds for the models incorporating NfL (Appendix 8.6; Figure 12a). The JLCMM yields lower Brier score values compared with the other two models for early predictions. However, over longer prediction horizons the differences between models reduces, with values being low, stable and around the same range as for models including p-tau181.

The same can be said for the time-dependent Brier score values among the models incorporating both biomarkers simultaneously. Appendix 8.6; Figure 12b demonstrates that the Cox PH model and the JM yield similar, stable AUC values across all prediction horizons, ranging from approximately 0.05 at for earlier predictions to 0.14 at later horizons.

Table 11 presents the Brier score values for each modelling approach and at each horizon.

Table 11: Time-dependent Brier score by model and prediction time point

Model	Brier_2y	Brier_3y	Brier_4y	Brier_5y	Brier_6y
Cox ptau-181	0.054	0.094	0.118	0.142	0.158
Cox NfL	0.054	0.098	0.122	0.145	0.159
Cox both	0.052	0.090	0.113	0.133	0.142
JM ptau-181	0.054	0.094	0.118	0.138	0.149
JM NfL	0.053	0.096	0.118	0.137	0.150
JM both	0.052	0.091	0.111	0.128	0.138
JLCMM ptau-181	0.024	0.072	0.104	0.131	0.144
JLCMM NfL	0.024	0.074	0.106	0.129	0.144

For predictions at two years, the JLCMM including NfL achieves the lowest Brier score, with a value of 0.024, compared with 0.053 for the JM including NfL and 0.054 for the Cox PH model including NfL. A similar pattern is observed for predictions using the p-tau181-based models at the same time point, where the JLCMM also yields the lowest Brier score, with a value of 0.024, compared with 0.055 for the JM and 0.054 for the Cox PH model.

5.3 Biomarker Comparison

The predictive performance of the models was also examined in order to compare the different biomarker specifications, meaning modelling either using p-tau181, NfL or a combination of the two biomarkers. Figure 4 illustrates how the time-dependent AUC of the models changes with biomarker specification across prediction horizons, with the model being fixed as Cox PH.

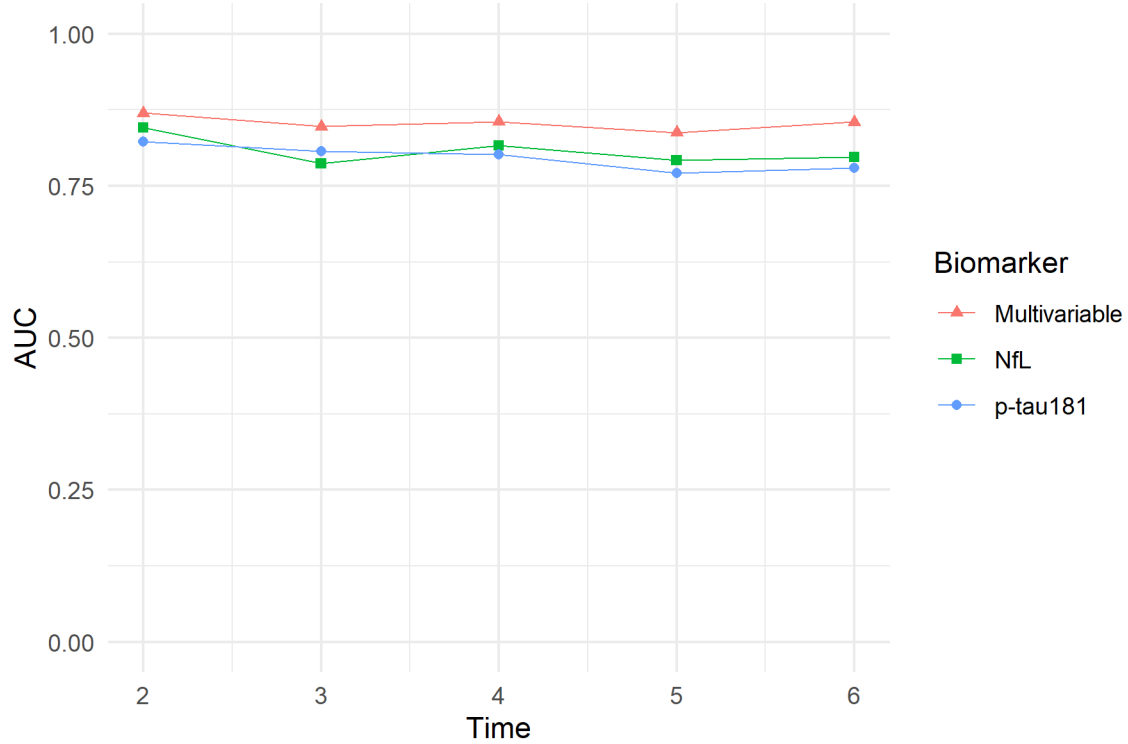


Figure 4: Time-dependent AUC (Cox PH)

Among the Cox PH models, the model incorporating NfL shows slightly higher time-dependent AUC values than the model incorporating p-tau181 across all horizons, with the exception of predictions made three years from baseline. However, both single-marker models are consistently outperformed by the combined model.

Within the JM framework, a similar overall pattern is observed. Appendix 8.7; Figure 13a demonstrates that the multivariate JM provides the strongest predictive performance across all horizons, with consistently higher AUC values than the univariate models. Nevertheless, the NfL-based model generally shows slightly higher values than the p-tau181-based model.

Comparing JLCMMs, NfL consistently shows slightly higher AUC values than the model including p-tau181, except for predictions made at the three-year point, where the p-tau181 model shows a slightly higher value (Appendix 8.7; Figure 13b).

The time-dependent AUC values for each biomarker specification at each horizon are also presented in Table 10. For example, for predictions made two years from baseline, the multivariate JM achieved an AUC value of 0.967, compared to 0.953 for the NfL model and 0.902 for the p-tau181 model. This indicates that the multivariate model provides the highest predictive performance at this time point.

However, when observing the time-dependent Brier score curves, such as Figure 5 presenting the curves for the Cox PH models, the difference in predictive performance between the models with different biomarker specifications becomes suddenly less evident.

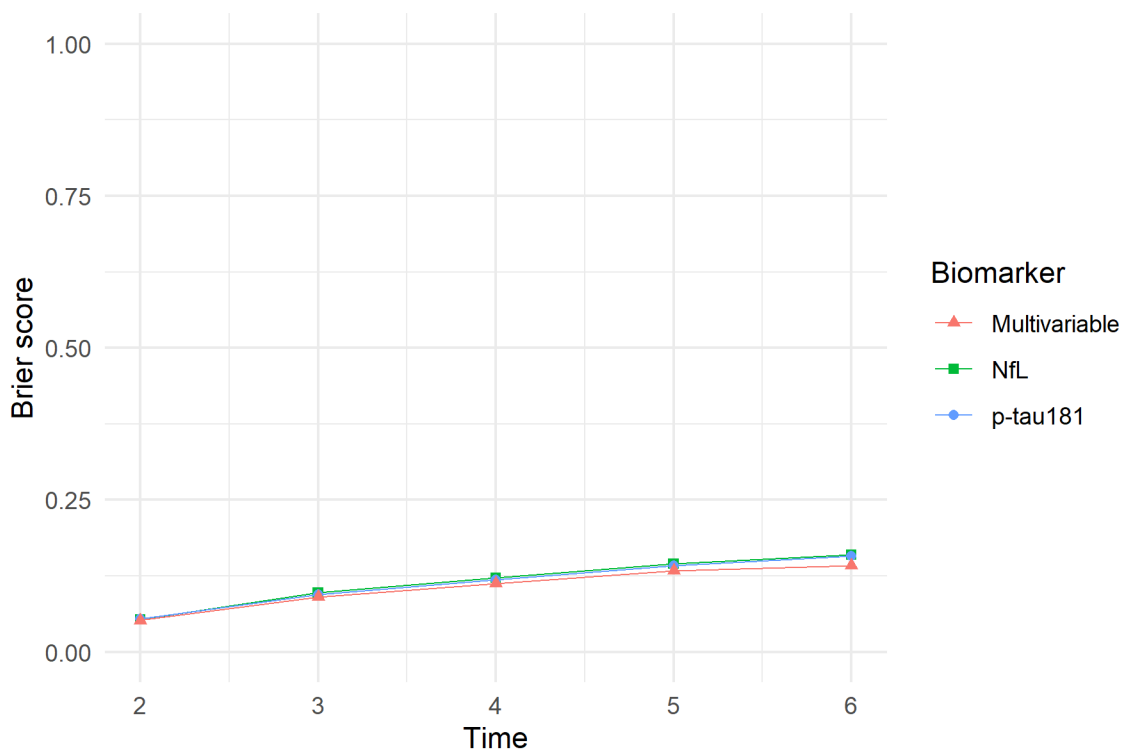


Figure 5: Time-Dependent Brier scores (Cox PH)

All three Cox PH models yield low and similar time-dependent Brier scores across all horizons, as reflected by the closely aligned curves. Small differences can be observed for predictions made at year six, although they remain minimal. A similar pattern is also observed among the JMs and JLCMMs (Appendix 8.7; Figure 14a-b).

Table 11, containing the time-dependent Brier score values for each biomarker specification and at each horizon, also confirms this pattern. For instance, for all three Cox PH models, predictions made two years ahead yield values ranging from 0.052 to 0.054, indicating minimal differences in predictive performance at this time point. For predictions made six years ahead, the differences between models becomes slightly more pronounced: modeling baseline values for both biomarkers yields a value of 0.149 at this horizon, whilst the models with baseline p-tau181 and NfL as covariates yield values of 0.158 and 0.159, respectively.

5.4 Sensitivity (or precision?) Analyses

To assure the robustness of these results, **sensitivity(?)** analyses were run.

In response to the debate over whether individuals with sparse data should be included when modeling longitudinal trajectories, all models were also estimated on an extended cohort of individuals with one or more p-tau181 and NfL measurements.

For the Cox PH models, the assumption of proportional hazards was also fulfilled for this cohort. The time-dependent AUC and Brier scores were in concordance with those from predictions made with the cohort restricted to those with 3 or more biomarker measurements.

To address the interval censored nature of the data, event times were also defined as the midpoint between the diagnosis exams when a positive dementia diagnosis was given.

The results from all **sensitivity(?)** analyses are available upon request.

6 Discussion

6.1 Main Findings and Conclusions

This study compared the predictive performance of different modeling frameworks for time-to-AD-dementia in individuals from the ADNI cohort, structured across three levels of information and their corresponding model approaches. At the most basic level, baseline biomarker information was used in Cox PH models estimated with either p-tau181, NfL or both jointly. The second level incorporated longitudinal biomarker information by extending these Cox PH models within a joint modeling framework, accounting for repeated measurements over time. The most complex level combined longitudinal biomarker trajectories with latent class structure using the JLCMM framework to capture heterogeneity in disease progression by modeling class-specific trajectories and survival processes. More precisely, two class models were fitted for p-tau181 and NfL measurements.

The predictive performance of these models was assessed by comparing time-dependent AUC and Brier scores with a landmark time of 1 year over horizons spanning 1-5 years. Both evaluations demonstrated that all three model approaches showed relatively good predictive performance, with AUC-values above 0.75 and Brier scores below approximately 0.15 across all horizons.

At the same time, the time-dependent AUC showed that the predictive performance differed across the three levels of information considered. Overall, incorporating longitudinal biomarker trajectories improved discriminatory ability compared to baseline only measurements. This improvement was consistent across biomarkers (p-tau181, NfL, and their combination) and was most pronounced for early predictions, where JMs achieved the highest AUC values relative to the Cox PH models. These findings are consistent with those of Goksuluk et al. (2025,

p. 14), who reported improved predictive performance of joint modelling approaches compared to Cox PH models using, among other specifications, baseline biomarker values. The difference between JMs and the other models were more pronounced at earlier time horizons, suggesting that JMs may offer the best early predictions. This is clinically meaningful, as early identification of those at risk of AD-dementia could be beneficial in clinical trial enrollment. However, JLCMMs achieved comparable or higher AUC values at intermediate to longer horizons. This indicates that there is no single modelling approach that consistently dominates in terms of time-dependent AUC across all time horizons, and even suggests a time-dependent shift in relative model performance.

The corresponding time-dependent Brier scores presented a broadly consistent but complementary pattern. JMs generally improved predictive accuracy compared to Cox PH models over intermediate and longer horizons, reflecting better overall calibration when longitudinal information was incorporated. However, the short-term predictive performance of the JLCMMs were superior when compared against the other models. This suggests different modelling frameworks capture distinct aspects of predictive performance. While JMs optimised short-term discriminatory ability, JLCMMs appeared to improve short-term absolute risk prediction and calibration. In other words, JMs performed better in correctly ranking individuals according to risk, while latent class structure contributed more to the accuracy of predicted probabilities in the short term.

Across all approaches, differences between the three levels of information were most evident at early follow-up, while performance became more similar at longer time horizons. This convergence was observed for both time-dependent AUC and time-dependent Brier score and indicates that biomarker levels and their early longitudinal changes are most informative for short-term prediction, while increasing uncertainty and unobserved factors reduce both discriminatory ability and calibration accuracy at longer follow-up periods.

Goksuluk et al. (2025, p. 11) reported that models including multiple biomarkers simultaneously performed slightly better than models including a single biomarker, which is also supported by the findings of the present study. Across all modelling approaches, the combination of p-tau181 and NfL provided the best overall predictive performance for progression to Alzheimer’s disease-related dementia compared to either biomarker alone. This was evident in both discrimination and calibration, although the magnitude of the difference was generally modest at short-term follow-up and became slightly more pronounced over longer prediction horizons, particularly for the Brier score.

More precisely, the combined biomarker specification yielded higher time-dependent AUC values and lower time-dependent Brier scores across all follow-up horizons. In contrast, p-tau181 and NfL individually showed broadly comparable performance without a consistent advantage of one biomarker over the other, indicating that neither biomarker alone fully captures the available prognostic information. Instead, the two biomarkers appear to provide complementary information, with p-tau181 reflecting Alzheimer’s disease-specific pathology and NfL capturing more general neurodegenerative processes. When combined, they improved both the ranking of individuals according to risk and the accuracy of absolute risk predictions.

Sensitivity analyses were made to ensure that these results were robust, more precisely by changing

Diagnostic exam dates were used as a proxy for AD onset in this analysis. Considering the exact time of dementia onset is unknown, individuals are interval censored. This has the potential to introduce bias in the estimated event times, as the time-to-diagnosis is overestimated. Despite this, interval censoring was not adequately accounted for given the level of this thesis and its time constraints. This would require more complex statistical methods, for which standard Cox PH models would no longer be directly applicable. As a direction for future research, it is therefore recommended that progression to dementia be modeled using interval censored survival methods in order to reduce potential bias and better reflect the underlying disease process.

Additionally, this analysis did not compare a multivariate JLCMM modeling both p-tau181 and NfL measurements, due to time constraints. Currently, the *lcmm* package does not offer dynamic predictions for multivariate JLCMMs, meaning there would be a heavier time burden were this model to be included. Therefore, further research could work to assess the benefit in class-specific modeling of these biomarkers simultaneously.

Finally, as plasma biomarker research is still emerging, rich longitudinal data sets for these biomarkers remain limited. For example, individuals in the data used for this analysis had at most five biomarker measurements. Future studies should therefore aim to replicate and extend this analysis once longitudinal biomarker data is available over longer time frames or cohorts with more frequent longitudinal sampling become available. This would improve the precision of both trajectory estimation and the assessment of associations between biomarker dynamics and disease progression.

7 Bibliography

Alzheimer's Association (n.d.). *Mild cognitive impairment (MCI)*. https://www.alz.org/alzheimers-dementia/what-is-dementia/related_conditions/mild-cognitive-impairment

Alzheimer's Disease Neuroimaging Initiative (ADNI) (n.d.-a). *About ADNI*. <https://adni.loni.usc.edu/about/>

- (n.d.-b). *ADNI Data*. <https://adni.loni.usc.edu/data-samples/adni-data/>

Alzheimer's Society (n.d.). *Risk factors for Alzheimer's disease*. <https://www.alzheimers.org.uk/about-dementia/types-dementia/who-gets-alzheimers-disease>

Biomarkers Definitions Working Group. (2001). Biomarkers and surrogate endpoints: preferred definitions and conceptual framework". *Clinical pharmacology and therapeutics*, 69(3), 89-95. <https://doi.org/10.1067/mcp.2001.113989>

- Brown, E. R., & Ibrahim, J. G. (2003). A Bayesian semiparametric joint hierarchical model for longitudinal and survival data. *Biometrics*, 59(2), 221-228. <https://doi.org/10.1111/1541-0420.00028>
- Carlin, B. P., & Louis, T. A. (2008). *Bayesian methods for data analysis* (3rd ed.). CRC Press.
- Carrillo, M. C., Dean, R. A., Nicolas, F., Miller, D. S., Berman, R., Khachaturian, Z., Bain, L. J., Schindler, R., Knopman, D., & Alzheimer's Association Research Roundtable (2013). Revisiting the framework of the National Institute on Aging-Alzheimer's Association diagnostic criteria. *Alzheimer's & dementia : the journal of the Alzheimer's Association*, 9(5), 594-601. <https://doi.org/10.1016/j.jalz.2013.05.1762>
- Celeux, G. & Soromenho, G. (1996). An entropy criterion for assessing the number of clusters in a mixture model. *Journal of Classification*: 13(1), 195-212. <https://doi.org/10.1007/BF01246098>
- Chang, L., Rissin, D. M., Fournier, D. R., Piech, T., Patel, P. P., Wilson, D. H., & Duffy, D. C. (2012). Single Molecule Enzyme-Linked Immunosorbent Assays: Theoretical Considerations. *Journal of Immunological Methods*, 378(1-2), 102-115. <https://doi.org/10.1016/j.jim.2012.02.011>
- Clark, C., Lewczuk, P., Kornhuber, J., Richiardi, J., Maréchal, B., Karikari, T. K., Blennow, K., Zetterberg, H., & Popp, J. (2021). Plasma neurofilament light and phosphorylated tau 181 as biomarkers of Alzheimer's disease pathology and clinical disease progression. *Alzheimer's Research & Therapy*, 13(1). <https://doi.org/10.1186/s13195-021-00805-8>
- Fitzmaurice, G. M., Laird, N. M., & Ware, J. H. (2011). *Applied longitudinal analysis* (2nd ed.). John Wiley & Sons.
- Goksuluk, M. B., Goksuluk, D., & Sipahioglu, M. H. (2025). Assessing biomarker trajectories for mortality risk in peritoneal dialysis: A focus on multivariate joint modeling. *PloS one*, 20(7), e0320385. <https://doi.org/10.1371/journal.pone.0320385>
- Grande, G., Valletta, M., Rizzuto, D., et al. (2025). Blood-based biomarkers of Alzheimer's disease and incident dementia in the community. *Nature Medicine*, 31(1), 2027-2035. <https://doi.org/10.1038/s41591-025-03605-x>
- Guo, G., Song, W., Wang, A., Cui, Q., Yang, X., Wang, Y., Ma, Y., Han, H., Li, Z., Zhang, Z., Meng, W., & Wang, S. (2026). Sensitivity comparison of longitudinal cognitive function indicators of Alzheimer's disease after mild cognitive impairment: a prospective cohort study. *Sci Rep*, 16(1), 14503. <https://doi.org/10.1038/s41598-026-44192-2>
- Hamra, G., MacLehose, R., & Richardson, D. (2013). Markov chain Monte Carlo: an introduction for epidemiologists. *International journal of epidemiology*, 42(2), 627-634. <https://doi.org/10.1093/ije/dyt043>
- Jack, C. R., Andrews, J. S., Beach, T. G., Buracchio, T., Dunn, B., Graf, A., Hansson, O., Ho, C., Jagust, W., McDade, E., Molinuevo, J. L., Okonkwo, O. C., Pani, L., Rafii,

- M. S., Scheltens, P., Siemers, E., Snyder, H. M., Sperling, R., Teunissen, C. E., & Carrillo, M. C. (2024). Revised criteria for diagnosis and staging of Alzheimer’s disease: Alzheimer’s Association Workgroup. *Alzheimer’s & Dementia*, 20(8). <https://doi.org/10.1002/alz.13859>
- Jacqmin-Gadda, H., Proust-Lima, C., Taylor, J. M. G., & Commenges, D. (2010). Score Test for Conditional Independence Between Longitudinal Outcome and Time to Event Given the Classes in the Joint Latent Class Model. *Biometrics*, 66(1), 11-19. <http://www.jstor.org/stable/40663147>
- Kamarudin, A. N., Cox, T., & Kolamunnage-Dona, R. (2017). Time-dependent ROC curve analysis in medical research: Current methods and applications. *BMC Medical Research Methodology*, 17(1), 53. <https://doi.org/10.1186/s12874-017-0332-6>
- Kleinbaum, D. G., & Klein, M. (2012). *Survival analysis: A self-learning text* (3rd ed.). Springer.
- Kyheng, M., Babykina, G., & Duhamel, A. (2025). Joint latent class models: A tutorial on practical applications in clinical research. *Statistics in Medicine*, 44(8-9), e70047. <https://doi.org/10.1002/sim.70047>
- Mattsson, N., Cullen, N. C., Andreasson, U., Zetterberg, H., & Blennow, K. (2019). Association Between Longitudinal Plasma Neurofilament Light and Neurodegeneration in Patients With Alzheimer Disease. *JAMA neurology*, 76(7), 791-799. <https://doi.org/10.1001/jamaneurol.2019.0765>
- Moore, D. F. (2016). Regression Analysis Using the Proportional Hazards Model. *Use R!*, 55-72. https://doi.org/10.1007/978-3-319-31245-3_5
- Park, K. M., Sung, J. M., Kim, W. J., An, S. K., Namkoong, K., Lee, E., & Chang, H. J. (2019). Population-based dementia prediction model using Korean public health examination data: A cohort study. *PloS one*, 14(2), e0211957. <https://doi.org/10.1371/journal.pone.0211957>
- Peng, R.E. (2022). *Advanced Statistical Computing*. <https://bookdown.org/rdpeng/advstatcomp/monitoring-convergence.html#gelman-rubin-statistic>
- Prince, T., Bommert, A., Rahnenführer, J., & Schmid, M. (2025). On the estimation of inverse-probability-of-censoring weights for the evaluation of survival prediction error. *PLOS ONE*, 20(1), e0318349. <https://doi.org/10.1371/journal.pone.0318349>
- Proust-Lima, C., Saulnier, T., Philipps, V., Traon, A. P., Péran, P., Rascol, O., Meissner, W. G., & Foubert-Samier, A. (2023). Describing complex disease progression using joint latent class models for multivariate longitudinal markers and clinical endpoints. *Statistics in medicine*, 42(22), 3996-4014. <https://doi.org/10.1002/sim.9844>
- Rissin, D. M., Kan, C. W., Campbell, T. G., Howes, S. C., Fournier, D. R., Song, L., Piech, T., Patel, P. P., Chang, L., Rivnak, A. J., Ferrell, E. P., Randall, J. D., Provuncher, G. K., Walt, D. R., & Duffy, D. C. (2010). Single-molecule enzyme-linked immunosorbent assay

detects serum proteins at subfemtomolar concentrations. *Nature Biotechnology*, 28(6), 595-599. <https://doi.org/10.1038/nbt.1641>

Rizopoulos, D. (2012). *Joint Models for Longitudinal and Time-to-Event Data: With Applications in R* (1st ed.). Chapman and Hall/CRC. <https://doi.org/10.1201/b12208>

- (2018). *Multivariate joint models*. <https://www.drizopoulos.com/vignettes/multivariate%20joint%20models>

Scollard, P., Mukherjee, S., Choi, S. E., Lee, M. L., Klinedinst, B., Gibbons, L. E., Trittschuh, E. H., Mez, J., Saykin, A. J., James, B. D., Proust-Lima, C., & Crane, P. K. (2026). Heterogeneous patterns of cognitive decline in Alzheimer's disease across three domains of cognition. *Journal of Alzheimer's disease : JAD*, 110(1), 383-396. <https://doi.org/10.1177/13872877251414975>

Park, S. Y., Park, J. E., Kim, H., & Park, S. H. (2021). Review of Statistical Methods for Evaluating the Performance of Survival or Other Time-to-Event Prediction Models (from Conventional to Deep Learning Approaches). *Korean journal of radiology*, 22(10), 1697-1707. <https://doi.org/10.3348/kjr.2021.0223>

Therneau, T.M. & Grambsch, P.M (2000). *Modeling Survival Data: Extending the Cox Model*. Springer, Berlin. <https://doi.org/10.1007/978-1-4757-3294-8>

Verbeke, G., & Molenberghs, G. (2000). *Linear mixed models for longitudinal data*. Springer Science & Business Media.

Vatcheva, K. P., Lee, M., McCormick, J. B., & Rahbar, M. H. (2016). Multicollinearity in Regression Analyses Conducted in Epidemiologic Studies. *Epidemiology (Sunnyvale, Calif.)*, 6(2), 227. <https://doi.org/10.4172/2161-1165.1000227>

Wang, Y. L., Chen, J., Du, Z. L., Weng, H., Zhang, Y., Li, R., Jia, Z., Sun, M., Jiang, J., Wang, F. Z., Xu, J., & Alzheimer's Disease Neuroimaging Initiative (2021). Plasma p-tau181 Level Predicts Neurodegeneration and Progression to Alzheimer's Dementia: A Longitudinal Study. *Frontiers in neurology*, 12(1), 695696. <https://doi.org/10.3389/fneur.2021.695696>

Wang, C., Shen, J., Charalambous, C., & Pan, J. (2024). Modeling biomarker variability in joint analysis of longitudinal and time-to-event data. *Biostatistics (Oxford, England)*, 25(2), 577-596. <https://doi.org/10.1093/biostatistics/kxad009>

Wong, P.C., Savonenko, A., Li, T., Price, DL. (2012). 'Neurobiology of Alzheimer's Disease' in Brady, S.T., Siegel, G.J., Albers, RW., Price, D.L (8 ed.) *Basic neurochemistry : principles of molecular, cellular, and medical neurobiology*. Waltham, Massachusetts; Oxford: Academic Press / Elsevier.

World Health Organization (2024). *Preferred product characteristics of blood-based biomarker diagnostics for Alzheimer disease*. <https://iris.who.int/server/api/core/bitstreams/43c83ef7-7a34-4b56-ae17-b584f2fc0798/content>

- (2025). *Dementia*. <https://www.who.int/news-room/fact-sheets/detal/dementia>

Wu, L., Liu, W., & Hu, X. J. (2010). Joint inference on HIV viral dynamics and immune suppression in presence of measurement errors. *Biometrics*, 66(2), 327-335. <https://doi.org/10.1111/j.1541-0420.2009.01308.x>

Xia, X., Duffner, L. A., Bintener, C., Bradshaw, A., Lamirel, D., & Jönsson, L. (2025). Diagnostic and prognostic multimodal prediction models in Alzheimer's disease: A scoping review. *Journal of Alzheimer's disease : JAD*, 108(1), 209-221. <https://doi.org/10.1177/13872877251351630>

Xu, L. (2021). Bayesian Multivariate Joint Modeling for Skewed-longitudinal and Time-to-event Data. *USF Tampa Graduate Theses and Dissertations*. <https://digitalcommons.usf.edu/etd/9268>

Yuan, M., Lian, S., Li, X., Long, X., Fang, Y., & Alzheimer's Disease Neuroimaging Initiative (ADNI) (2024). Blood biomarkers in dynamic prediction of conversion to Alzheimer's disease: An application of joint modeling. *International journal of geriatric psychiatry*, 39(3), e6079. <https://doi.org/10.1002/gps.6079>

Zou, K. H., Liu, A., Bandos, A. I., Ohno-Machado, L., & Rockette, H. E. (2012). *Statistical evaluation of diagnostic performance: Topics in ROC analysis* (1st ed.). Chapman and Hall/CRC. <https://doi.org/10.1201/b11031>

8 Appendix

8.1 Descriptive Statistics

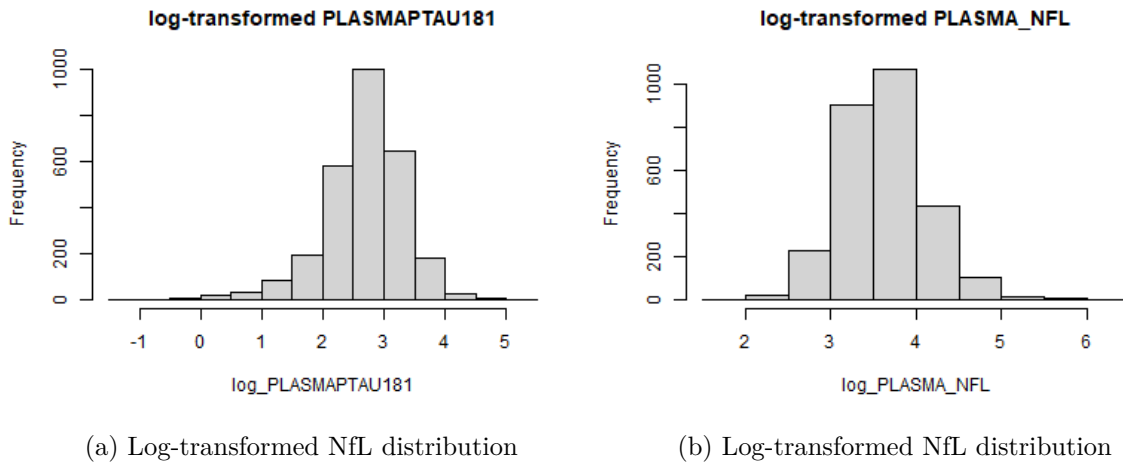
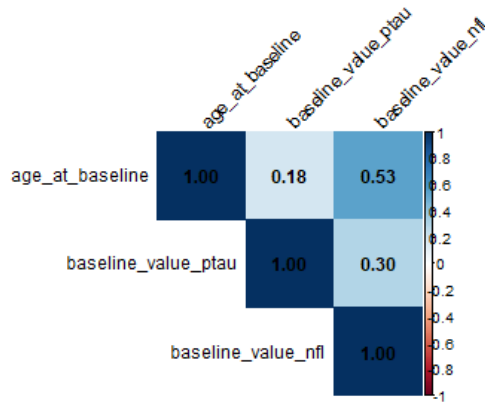
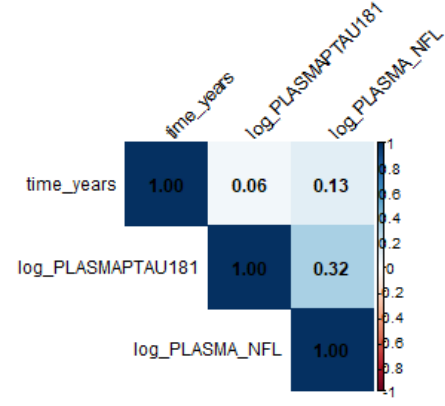


Figure 6: Distributions of log transformed biomarkers



(a) Variables in the Cox PH models



(b) Variables in the LME models

Figure 7: Correlation plots of numerical variables

8.2 Test Results

Table 12: Results of the likelihood ratio tests to determine the random effects structure for LME models

Model	Test statistic	P-value
ptau-181	1.488	0.475
NfL	10.506	0.005

Table 13: p-values for the proportional hazards assumption tests for all covariates in the Cox PH models

variable	Standard	p-tau181 baseline	NfL baseline	Both baseline
age_at_baseline	0.002	0.002	0.003	0.002
PTGENDER	0.803	0.835	0.827	0.888
APOE4_status	0.158	0.173	0.127	0.154
GLOBAL	0.013	0.028	0.031	0.057
baseline_value_ptau	NA	0.698	NA	0.632
baseline_value_nfl	NA	NA	0.140	0.113

Table 14: Results of the likelihood ratio test for the Cox PH models with and without gender as a covariate

Model	Test statistic	P-value
ptau-181 baseline	0.0006607	0.98
NfL baseline	0.0037223	0.95
ptau-181 and NfL baseline	0.1184134	0.73

8.3 Residual Diagnostics

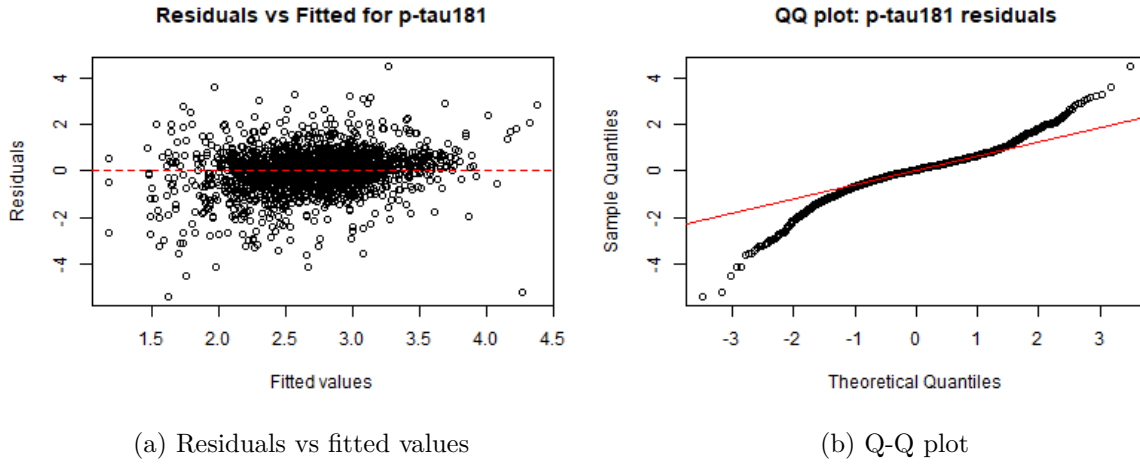


Figure 8: LME model with longitudinal p-tau181 measurements

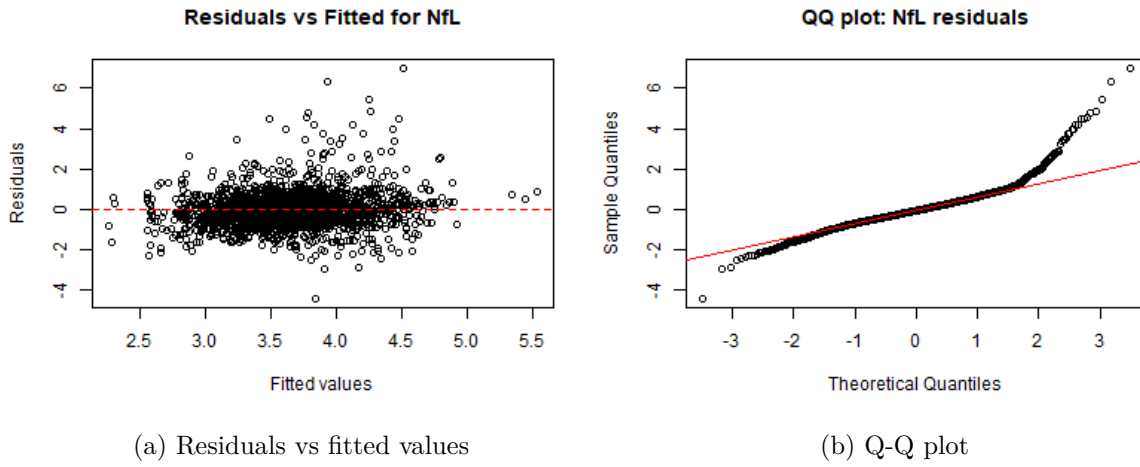


Figure 9: LME model with longitudinal NFL measurements

8.4 Convergence

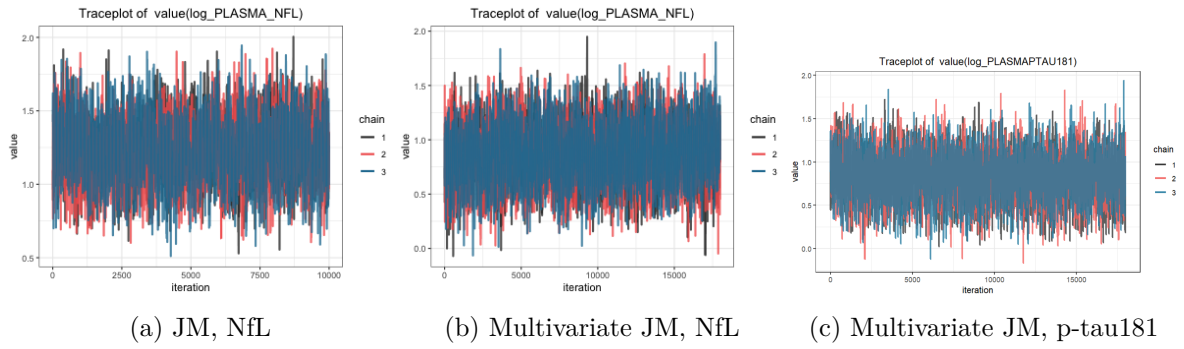


Figure 10: Trace plots of the α -parameters (convergence)

Table 15: Model parameters for the two class JLCMM with longitudinal measurements for p-tau181

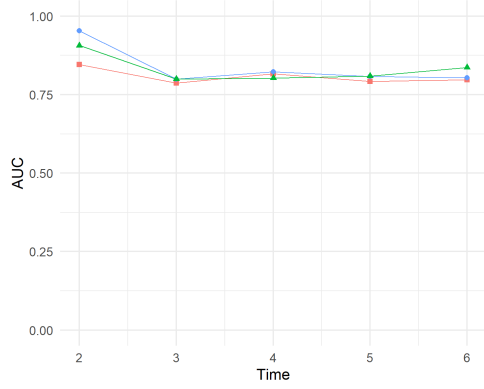
Submodel	Parameter	Estimate	SE	Z	P value
Class Membership	intercept class1	-0.720	0.176	-4.087	0
Survival	event1 +/-sqrt(Weibull1) class 1	0.453	0.033	13.608	0
Survival	event1 +/-sqrt(Weibull2) class 1	1.216	0.063	19.188	0
Survival	event1 +/-sqrt(Weibull1) class 2	0.231	0.012	18.937	0
Survival	event1 +/-sqrt(Weibull2) class 2	2.746	0.474	5.795	0
Longitudinal	intercept class1	2.925	0.044	66.992	0
Longitudinal	intercept class2	2.551	0.035	72.324	0
Longitudinal	time_years	0.028	0.007	4.183	0

Table 16: Model parameters for the two class JLCMM with longitudinal measurements for NFL

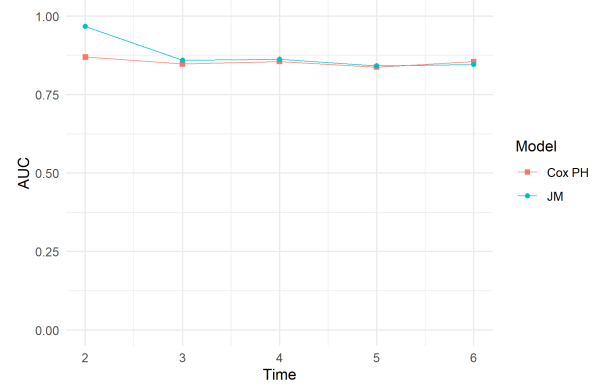
Submodel	Parameter	Estimate	SE	Z	P value
Class Membership	intercept class1	0.809	0.145	5.588	0
Survival	event1 +/-sqrt(Weibull1) class 1	0.232	0.012	19.118	0
Survival	event1 +/-sqrt(Weibull2) class 1	2.681	0.476	5.638	0
Survival	event1 +/-sqrt(Weibull1) class 2	0.471	0.026	18.063	0
Survival	event1 +/-sqrt(Weibull2) class 2	1.239	0.058	21.320	0
Longitudinal	intercept class1	3.458	0.026	130.853	0
Longitudinal	intercept class2	3.766	0.039	96.638	0
Longitudinal	time_years	0.045	0.004	11.971	0

8.5 Model Results

8.6 AUC and Brier Curves: Model Comparison

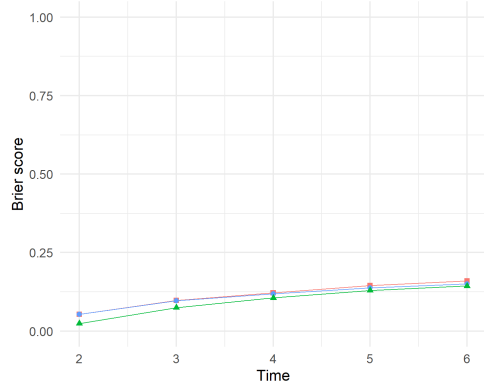


(a) Models using NfL

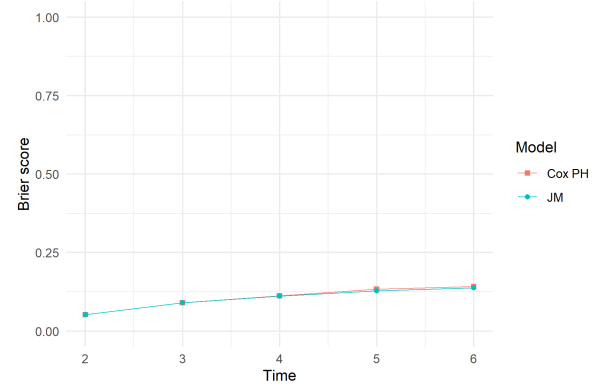


(b) Models using p-tau181 & NfL

Figure 11: Time-dependent AUC curves



(a) Models using NfL



(b) Models using p-tau181 & NfL

Figure 12: Time-dependent Brier score curves

8.7 AUC and Brier Curves: Biomarker Comparison

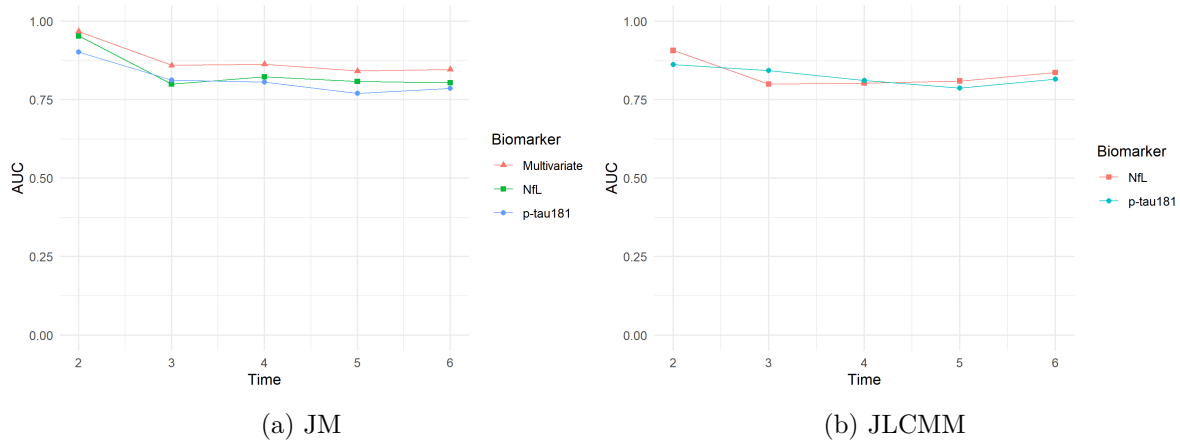


Figure 13: Time-dependent AUC curves

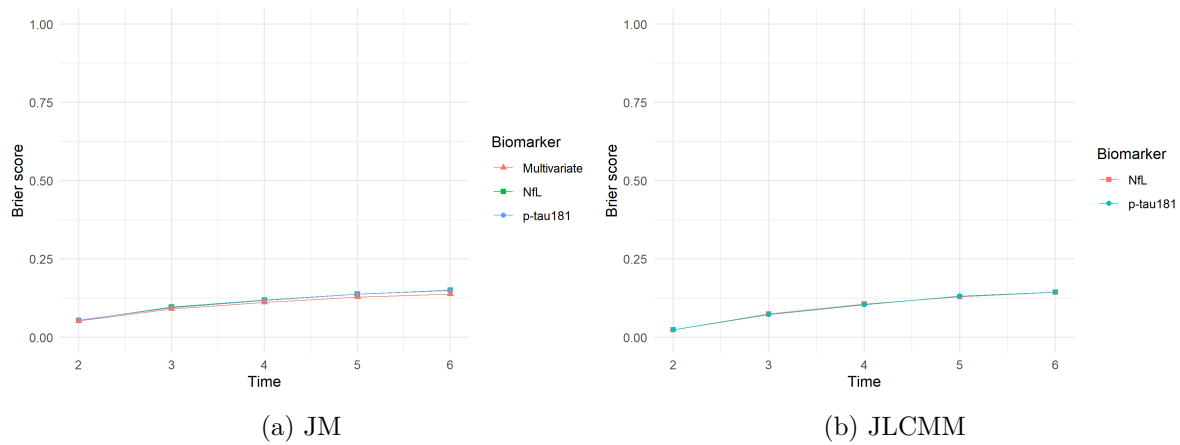


Figure 14: Time-dependent Brier score curves

8.8 Sensitivity analysis: 1+ Biomarker Measurements

8.9 Results of sensitivity analyses

Table 17: Time-depending Brier scores when visits with duplicate measurements are replaced by visits with the last measurement

Model	AUC_2y	AUC_3y	AUC_4y	AUC_5y	AUC_6y
Cox ptau-181	0.824	0.751	0.789	0.773	0.757
Cox NfL	0.801	0.739	0.767	0.769	0.778
Cox both	0.826	0.768	0.800	0.806	0.808
JM ptau-181	0.725	0.738	0.782	0.762	0.742
JM NfL	0.594	0.667	0.728	0.747	0.756
JM both	0.692	0.735	0.781	0.787	0.783
JLCMM ptau-181	0.645	0.725	0.745	0.749	0.729
JLCMM NfL	0.558	0.624	0.631	0.685	0.710